

PAGE: PARTITION-AWARE GATED KV-CACHE EVICTION

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ABSTRACT

KV-cache eviction can do more than compress. In long-context LLMs, keeping only some cached tokens sometimes matches or exceeds full-cache accuracy, because many redundant prefill tokens otherwise dilute attention away from the tokens that carry the answer. This benefit is not uniform: evicting the wrong tokens drops accuracy to zero on tasks that need precise retrieval. We show that one label-free number computed from the prefill attention, the drop between early and late layers in how much attention heads agree on which tokens to read, predicts per input, before any decoding, which of the two regimes an input is in. We build this into PAGE (Partition-Aware Gated Eviction), a wrapper that runs any SnapKV-style evictor when the drop is large and keeps the full cache when it is small, with no training, labels, or fine-tuning. PAGE is a safety mechanism: on inputs where eviction would be catastrophic it keeps the full cache, and elsewhere it passes the base evictor through unchanged. On multi-key retrieval with Mistral-7B, plain SnapKV falls from 99% to 0% as the budget shrinks while PAGE holds it at 89%. On the accuracy–memory frontier, across four base evictors (SnapKV, H2O, StreamingLLM, PyramidKV) and four models, PAGE improves on plain eviction by 3 to 19pp at equal achieved cache whenever the target keeps more than about a third of the cache, and by design defers to plain eviction under extreme compression. The head-agreement drop orders tasks the same way across all four model families; one fixed threshold works on the Qwen2.5 and Mistral families, and an unlabeled per-model pilot extends it to Llama-family and Qwen3 models. We frame when eviction helps and when it hurts as a task-type partition, dilution-prone versus capacity-bound, and give a scaling relation for the fraction of inputs that recover, stress-tested out of range at 32K. Code and per-input logs are released.

1 INTRODUCTION

Long-context LLMs spend most of their inference cost on the KV-cache, and a sequence of recent papers shows that eviction can do more than compress. Bui et al. (2026) prove that selective retention can reduce a quantifiable *attention dilution* term and report a learned-gate evictor (DB-TrimKV) that exceeds the full cache on long-context dialogue by several points. Yang et al. (2026a) derive a closed-form information-bottleneck score that produces the same paradox on LongBench; an indexer-and-latent-memory architecture (Yang et al., 2026c) reports an information-density peak on RULER and LongBench at moderate compression. The common claim across these papers is that less compute can yield more accuracy.

Open question. Each of these results reports the effect averaged over a benchmark suite where it occurs. None addresses two natural follow-up questions:

- (Q1) *For which inputs and tasks does eviction help, and for which does it hurt?* Mean accuracy hides per-input variance: a benchmark mean that is positive in expectation can mask a subset of inputs on which eviction is catastrophic.
- (Q2) *Can the partition be predicted a priori from cheap inference-time signals?* Existing methods that adapt either train a scorer (Bui et al., 2026; Yang et al., 2026c; Dong et al., 2026; Moschella et al., 2026) or derive an unconditional capacity score from query statistics

(Yang et al., 2026a). Whether the dilution-prone vs. capacity-bound distinction is visible in the pre-eviction attention alone is open.

We answer Q1 and Q2 affirmatively. The *head-agreement drop* on prefill attention, the difference between mean Jaccard top- k head agreement in early vs. late layer-bins, partitions inputs into a dilution-prone class $\mathcal{D}$ and a capacity-bound class $\mathcal{C}$, and a drop-in gating wrapper turns the partition into a training-free policy that strictly improves any SnapKV-style evictor at matched budget.

Contributions.

1. **An empirical partition** (Section 3, Table 1). Across eight (model, context) cells and six RULER task families, precise multi-key needle-in-a-haystack retrieval with three keys (NIAH-MK3) is the one task where eviction never recovers accuracy, while the five dilution-prone families do recover wherever the model has headroom. A controlled distractor sweep confirms the mechanism: the head-agreement drop falls monotonically as near-tie distractors are added, separating the two classes without overlap (Figure 1).
2. **A label-free predictor** (Section 4.1). The early-to-late head-agreement drop $D := \text{agT}_{\text{early}} - \text{agT}_{\text{late}}$ orders tasks so that the capacity-bound one is smallest (Figure 1). One fixed threshold works on seven of eight Qwen/Mistral cells; a 40-input recipe fixes the eighth. The threshold does *not* transfer as a constant to other families (it fails outright on Qwen3), but the ordering does, so an unlabeled per-model standardization pilot recovers a working threshold on the Llama family and Qwen3 (Section 5).
3. **PAGE, a gating algorithm** (Section 4). One conditional on the prefill attention, with no training, back-propagation, or labels. It works as a safety mechanism: it keeps the full cache on inputs where eviction is catastrophic and passes the base evictor through otherwise. On the accuracy-memory frontier it improves on plain eviction by 3 to 19pp at equal achieved cache for targets above $\approx 1/3$ of full KV (Figure 2), and defers to plain eviction under extreme compression; the effect holds across four base evictors and four models (Table 2) and lifts SnapKV on two further Llama-family models.
4. **Head-to-heads with published methods** (Section 7.6). Wrapped around a CapKV re-implementation and the pretrained DBTrimKV evictor, the gate improves matched-budget accuracy, with the gain concentrated where the base method would otherwise collapse (and, on DBTrimKV, dominated by the full-cache fallback, which we quantify).
5. **A scaling law** (Section 6). For dilution-prone tasks, the fraction of recovering inputs grows with accuracy headroom and falls as the model saturates the task; the closed form $\rho \approx (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}/\gamma)$ (Section 6) unifies context-length amplification, model-size saturation, and the re-entry of larger models into the dilution regime at long context. We stress-test it with a pre-registered 32K extrapolation (Section 7.2): the directional content survives, but constant-ratio proportionality is falsified (1 of 3 predictions strictly inside its interval), so we report this as a scaling *relation*, not a quantitative law.

We do not claim as novel the paradigm that less compute can beat full attention, or the attention-dilution mechanism: Bui et al. (2026) state both, with a formal mechanism (their Proposition 3.1 and Corollary 3.2) that we adopt. Nor do we claim to be first to observe that eviction degrades some task types (Chen et al., 2026), that task type modulates which KV positions matter (Zhou et al., 2025), that one can ask when value-aware eviction helps (Zhang et al., 2026), or that head consensus relates to compression tolerance (Ananthanarayanan et al., 2026). Our contribution is the *conjunction* these works do not form: a joint helps-versus-hurts partition on a single axis, predicted a priori per input from one prefill scalar, deployed as a whether-to-evict gate, together with the recovery-fraction scaling relation. We also do not claim a new attention statistic; we repurpose a known head-agreement primitive (early-minus-late layers) as the gating signal.

2 RELATED WORK

The KV-eviction methods nearest to ours motivate it but differ in kind. DBTrimKV (Bui et al., 2026), whose single-head dilution mechanism we adopt, CapKV (Yang et al., 2026a), DynamicKV (Zhou et al., 2025), Protection-Is-All-You-Need (Garcia, 2026), and IndexMem (Yang et al., 2026c) all

decide, on every input, *which* tokens to keep under a budget. Diagnostic work documents that eviction degrades some task types (Chen et al., 2026), and value-state analyses ask when eviction helps (Zhang et al., 2026); neither pairs a task class where eviction helps with one where it hurts on the same axis, nor predicts the side a priori, and Zhang et al. (2026) keys on value states rather than the attention topology we use. Our gating signal, an early-to-late *cross-layer* drop in head agreement, is related to but distinct from the *within-layer* head-consensus metric that Ananthanarayanan et al. (2026) compute to explain compression tolerance descriptively; ours is a directional cross-layer contrast, computed online at prefill and thresholded to gate. Plug-and-play wrappers over several base evictors exist (VECTOR Lin et al., 2026b, CriticalKV Feng et al., 2026), and learned gates share our “gate” terminology (Attention-Gate Anonymous, 2024, Fast KVzip Kim et al., 2026), but all of these always compress and adapt *which* tokens are kept, or require training, whereas ours is training-free and decides *whether* to evict at all. Transfer of an eviction control across model families is itself known: CompilerKV (Yang et al., 2026b) ports compiled per-head retention tables across backbones; we instead transfer a single threshold, fixed across the Qwen and Mistral families and extended to Llama-family and Qwen3 models by an unlabeled per-model standardization of one scalar. The training-free base evictors we wrap are SnapKV (Li et al., 2024), H2O (Zhang et al., 2023), StreamingLLM (Xiao et al., 2024), and PyramidKV (Cai et al., 2025). A full treatment of all neighbors, including the DBTrimKV/CapKV head-to-heads and the many further concurrent evictors and allocators, is deferred to Appendix A.

3 THE EMPIRICAL PARTITION

Setup. We use RULER (Hsieh et al., 2024) via `simonjegou/ruler` (Jégou, 2024) at context lengths $T \in \{4K, 16K\}$ on four models: Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct, Qwen2.5-14B-Instruct (Yang et al., 2025), and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023). Eviction is SnapKV-style by default: each prompt position is scored by the average attention from the last $w = 32$ queries across heads and layers; we keep the top- b positions with $b \in \{0.0625, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875, 1.0\}$, and always preserve the first $n_{\text{sink}} = 4$ sink positions and the last $w = 32$ recent positions. RoPE alignment is preserved by cache slicing with explicit `position_ids` rather than the attention-mask trick which silently shifts RoPE for evicted positions. Multi-element answers (VT, FWE, `niah_multivalue`) are scored by RULER’s `string_match_all`; single-element answers (NIAH, QA) by `string_match_part`.

Metric. For task T , model M , and context length T we report the per-input recovery rate

$$\rho_{KV}(T, M, T) := \Pr_x[A(b^*(x), x) > A(b_{\max}, x)], \quad b^*(x) := \arg \max_b A(b, x). \quad (1)$$

This is the per-input strict-Pareto improvement frequency, a stronger statement than “mean accuracy is higher at some b .” We pre-register the threshold $\rho \geq 0.05$ as the operational definition of “dilution-prone” before any sweep is run.

The partition. Table 1 reports ρ on six RULER tasks across eight (model, context) cells, all under attention-score-based eviction. The result is qualitatively the same in every cell: five task families clear $\rho \geq 0.05$ on ≥ 1 context length (unless model saturation drives $1 - A_{\text{full}} \rightarrow 0$ and the scaling formula predicts $\rho \approx 0$ trivially, or the residual errors are budget-insensitive, as for QA_2 on Qwen2.5-14B below), and NIAH-MK3 has $\rho \leq 0.02$ on every model and every context (a 32K extension measures 0.04, Wilson [0.016, 0.098]; Section 7.2). Larger models at short context partially saturate (Qwen2.5-3B 4K has $\rho \approx 0$ on VT because $A_{\text{full}} \approx 1$), but the same models recover the effect at longer context (Qwen2.5-3B 16K FWE has $\rho = 0.32$). Mistral-7B shows the same shape: at 4K it is largely saturated ($A_{\text{full}} = 1.00$ on VT, 0.82 on FWE), and at 16K VT and FWE recover strongly ($\rho = 0.32$ and 0.22, and `niah_multivalue` 0.06 $\rightarrow$ 0.21). The QA family on Mistral, however, is budget-insensitive at both contexts (QA_1: $\rho = 0.04/0.00$ despite $A_{\text{full}} = 0.81/0.70$; QA_2: $\rho = 0.01/0.00$ despite $A_{\text{full}} = 0.50/0.56$), joining QA_2 on Qwen-14B: headroom exists but the wrong inputs stay wrong at every budget. All three exceptions are QA cells, which suggests parametric-knowledge failures that no cache budget can fix.

The detailed re-entry analysis of Qwen2.5-14B at long context and the demonstration that the partition is a near-tie-distractor structure rather than a task-name heuristic (including the non-RULER LongBench passage_retrieval_en probe) are deferred to Appendix A.

Table 1: Per-input recovery rate ρ on RULER with Wilson 95% intervals (small type). Bold marks point estimates $\rho \geq 0.05$. NIAH-MK3 (precise multi-key retrieval) is the only task that systematically fails the threshold in every cell tested. $N = 100$ per (task, cell) except Qwen-14B ($N = 50$ at 4K, $N = 30$ at 16K), Mistral 16K VT/FWE/QA.1/NIAH-MK3 ($N = 50$), Qwen-3B 16K NIAH-MK3 ($N = 50$), and Qwen-1.5B 16K NIAH-MK3 ($N = 46$). At these N most single cells straddle the 0.05 threshold and no per-cell call is individually powered; the claim rests on the pooled contrast, which is unambiguous (NIAH-MK3: $2/526$, $\rho = 0.004$, Wilson $[0.001, 0.014]$).

Task	Qwen 1.5B 4K	Qwen 1.5B 16K	Qwen 3B 4K	Qwen 3B 16K	Qwen 14B 4K	Qwen 14B 16K	Mistral 4K	Mistral 16K
VT	0.06 [.03, .13]	0.19 [.13, .28]	0.00 [.00, .04]	0.05 [.02, .11]	0.00 [.00, .07]	0.03 [.01, .17]	0.00 [.00, .04]	0.32 [.21, .46]
FWE	0.06 [.03, .13]	0.04 [.02, .10]	0.01 [.00, .05]	0.32 [.24, .42]	0.00 [.00, .07]	0.20 [.10, .37]	0.02 [.01, .07]	0.22 [.13, .35]
QA.1	0.08 [.04, .15]	0.14 [.09, .22]	0.03 [.01, .09]	0.08 [.04, .15]	0.06 [.02, .16]	0.10 [.03, .26]	0.04 [.02, .10]	0.00 [.00, .07]
QA.2	0.08 [.04, .15]	0.05 [.02, .11]	0.04 [.02, .10]	0.02 [.01, .07]	0.02 [.00, .11]	0.03 [.01, .17]	0.01 [.00, .05]	0.00 [.00, .04]
niah_multivalue	0.07 [.03, .14]	0.10 [.06, .17]	0.02 [.01, .07]	0.13 [.08, .21]	0.08 [.03, .19]	0.07 [.02, .21]	0.06 [.03, .13]	0.21 [.14, .30]
NIAH-MK3	0.01 [.00, .05]	0.00 [.00, .08]	0.00 [.00, .04]	0.02 [.00, .11]	0.00 [.00, .07]	0.00 [.00, .11]	0.00 [.00, .04]	0.00 [.00, .07]

Mechanism validation: distractor count drives D . A controlled sweep over the RULER NIAH-MultiKey family (Qwen2.5-1.5B, RULER 4K, $N = 50$ per task) confirms the dilution mechanism of Bui et al. (2026): mean head-agreement drop D shrinks monotonically as the near-tie distractor count grows, and the single-near-tie MK2 boundary case is analyzed alongside it. Full details and Table 3 are deferred to Appendix A.

The partition survives Garcia-scale protection. Re-running the Qwen2.5-1.5B 4K partition tasks with the 10% prefix + 10% suffix protection of Garcia (2026) leaves the partition unchanged: NIAH-MK3 still collapses and the eviction-helps signal survives on the dilution-prone side. Full numbers and caveats are deferred to Appendix A.

Random-eviction control. To rule out “any eviction helps” as an artifact, we replace SnapKV scoring with uniform-random scoring at every budget on the dilution-prone tasks. Random eviction destroys accuracy uniformly, ruling out a generic regularization explanation. Detailed numbers in Appendix B.

4 THE GATING METHOD: PAGE

4.1 ALGORITHM

PAGE (Partition-Aware Gated Eviction) wraps any SnapKV-style evictor in one conditional. Given a prompt x , the base evictor’s budget b , a sink-and-recency protection (n_{sink}, w) , and a threshold τ :

$$PAGE(x; \text{base}, b, \tau) = \begin{cases} \text{base.evict}(\text{cache}, b) & \text{if } D(\text{cache.attentions}) \geq \tau \\ \text{cache} & \text{otherwise} \end{cases} \quad (2)$$

where the prefill produces `cache` together with its prefill attentions and the head-agreement drop D is computed as in Eq. (3) below. The gate fires once per input, at prefill, before any decode step. No backward pass, no labels, no training, no per-token decision.

Head-agreement drop. For a prefill attention tensor of shape $[L, H, T, T]$ (L layers, H heads), we project onto the w observation queries (the last w rows) and average across them to get a per-(layer, head, key) attention vector. For each (layer, head pair (i, j)) we compute the Jaccard top- k similarity of the two heads’ top- k key sets (we use $k = 32$), and average over head pairs. This gives a per-layer head agreement a_ℓ . Binning layers into early third, middle third, late third, we set

$$D := a_{\text{early}} - a_{\text{late}}. \quad (3)$$

D is one scalar per input. It needs the per-head prefill attention over the w observation queries, computed at $O(LH^2k)$ cost, measured at 17ms on Qwen2.5-1.5B and 89ms on Mistral-7B at 4K (Section 7.5). This grows with head count and is not free even when the base evictor scores by attention, because that scoring uses head-averaged attention whereas D needs the per-head pattern; obtaining it requires materializing attention, which fused FlashAttention does not do (we discuss the deployment implication in Section 8).

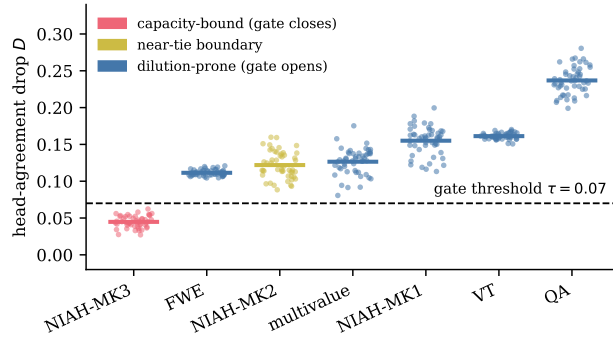


Figure 1: The head-agreement drop D separates the two task classes (Qwen2.5-1.5B, RULER 4K, $N = 50$ per task; each dot is one input, bars are per-task means). NIAH-MK3, the capacity-bound task, sits below the gate threshold $\tau = 0.07$; the dilation-prone tasks sit above it. The near-tie NIAH-MK1/MK2 tasks fall on the boundary, tracking the distractor-count gradient (App. A). The gate closes below the line (keep the full cache) and opens above it (run the base evictor).

Table 2: Headline 4×4 method-agnostic matrix. Same gate ($D \geq \tau$, $\tau = 0.07$), same calibration, no per-base-evictor tuning. Every one of 16 cells is positive; grand mean across the matrix is +22.9pp. The Qwen2.5-3B SnapKV cell was executed at $\tau = 0.04$ and is reported here at $\tau = 0.07$ via exact post-hoc re-evaluation (Appendix H); all other cells were executed at $\tau = 0.07$.

Model	SnapKV		H2O		StreamingLLM		PyramidKV	
	plain	gated	plain	gated	plain	gated	plain	gated
Qwen2.5-1.5B	0.438	0.559	0.196	0.357	0.033	0.195	0.417	0.570
Δ	+0.121		+0.162		+0.163		+0.152	
Qwen2.5-3B	0.580	0.839	0.339	0.642	0.052	0.482	0.536	0.865
Δ	+0.259		+0.303		+0.430		+0.329	
Qwen2.5-14B	0.709	0.853	0.668	0.868	0.080	0.330	0.632	0.860
Δ	+0.143		+0.200		+0.250		+0.228	
Mistral-7B	0.623	0.810	0.400	0.636	0.102	0.365	0.593	0.823
Δ	+0.187		+0.236		+0.263		+0.231	

Why this statistic? Figure 1 shows D separating the capacity-bound anchor from the dilation-prone tasks at the single threshold the gate uses. A same-inputs ablation against five other cheap prefill statistics shows that D is the only one that separates the capacity-bound anchor at all; every alternative fails task-level separation (per-input AUC up to 0.805 vs. D 's 1.000 on this Qwen2.5-1.5B cell, though D 's per-input separation drops to ≈ 0.80 on the Llama family; Section 8). The full ablation and Table 4 are deferred to Appendix A.

4.2 THE GATE IS ORTHOGONAL TO BASE-EVICTOR CHOICE

Table 2 reports the 4×4 matrix of four base evictors (SnapKV, H2O, StreamingLLM, PyramidKV) and four models, same gate, same $\tau = 0.07$, no per-method tuning. Each Δ is gated minus plain accuracy at matched *nominal* budget over eight budgets on the mixed suite. This is not a compression win: the gated arm holds more cache than plain at the same nominal budget, and almost all of Δ is the gate protecting NIAH-MK3 by refusing to evict it (Section 7.3 gives the matched-cache comparison). The matrix shows the *gate transfers across base evictors*, not that it advances the frontier at matched budget.

Cross-architecture SnapKV-only runs on two Llama-family models (Yi-1.5-9B, Llama-3.1-8B; Table 5), a scale-validation run at 32B, a best-plain-vs-best-gated comparison (Table 6) that rules out

“you only beat the weakest baseline,” and a discussion of why the deltas vary across base evictors are all deferred to Appendix A.

The clearest single result is on Mistral 4K NIAH-MK3, where the gate holds accuracy flat as plain SnapKV collapses to zero (Figure 3, quantified in Section 7).

A bar-chart rendering of the headline matrix (Figure 4) and a composition test against a fifth, geometry-based (ManifoldKV-style) base evictor, which the same gate lifts by $\Delta = +16.2\text{pp}$ and which shows composition extends outside the attention-score family, are deferred to Appendix A.

5 CALIBRATION RECIPE

The fixed $\tau = 0.07$ mis-classifies only one of the eight Qwen/Mistral (model, context) cells (Qwen2.5-3B 16K, where layer-bin drops compress at long context). A one-shot, label-free 40-input recipe, which picks 20 inputs from a capacity-bound task and 20 from a dilution-prone task and sets τ to the midpoint of their mean drops (Eq. (5)), fixes that failure cell, turning $+3.4\text{pp}$ into $+5.4\text{pp}$. An architecture-normalized variant needs no task labels: standardizing the drop per model (z -scoring over an unlabeled ~ 100 -input pilot) transfers a single global threshold zero-shot to the two held-out Llama-family models (Yi-1.5-9B, Llama-3.1-8B) and to Qwen3-4B, where the fixed τ fails outright. Full recipe, τ -sensitivity analysis, and the per-family z -score results (Table 7, Figure 5) are deferred to Appendix A.

6 THEORY: THE MECHANISM AND THE SCALING FORMULA

We adopt verbatim the dilution mechanism from Bui et al. (2026), which suffices to motivate our gating predictor: for a single attention head reading a relevant set $\mathcal{R}$ and a noise set $\mathcal{N}$ with pre-eviction signal mass $\alpha_{\mathcal{R}}$, near-tie distractors force the dilution δ_t (the attention mass outside the useful set) to be large, and preferential retention reduces it. The empirical partition follows: capacity-bound tasks concentrate a single signal ($\alpha_{\mathcal{R}} \rightarrow 1$), while dilution-prone tasks spread it ($\alpha_{\mathcal{R}} \ll 1$). Combining the head-SNR argument with the accuracy headroom $1 - A_{\text{full}}(M, T)$, the per-input expected accuracy improvement at a single budget b obeys the context-length scaling formula

$$\rho_{\text{KV}}(T, M, T) \approx (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}/\gamma), \quad p_{\text{recov}} : [0, \infty) \rightarrow [0, 1], \quad (4)$$

where $\rho_{\mathcal{R}}$ is the retained fraction of relevant mass and γ the noise-retention ratio. The full mechanism setup, the derivation that the partition follows from it, the context-length-scaling argument and its informal proposition, and the sufficient-condition bridge from the observable head-agreement drop D (Eq. (3)) to the dilution δ_t (Eq. (7)) are deferred to Appendix A, with the two proof sketches in Appendices E and F.

7 EXPERIMENTS

7.1 SETUP

Models, contexts, eviction policies, and protection settings are as in Section 3. The matrix uses RULER 4K with the four-task mixed suite (NIAH-MK3 + VT + FWE + QA_1, $N = 100$ each), as RULER 4K fits in one A100 80GB at all four base evictors and four models. The scaling table uses 4K and 16K. The gate cost is reported on Qwen2.5-1.5B and Mistral-7B at 4K; it scales super-linearly in head count ($O(H^2)$, Section 7.5).

Uncertainty quantification. All gating deltas carry 95% cluster-bootstrap confidence intervals over inputs and all recovery rates ρ Wilson score intervals; every one of the 16 matrix deltas is positive with its entire interval above zero (smallest lower bound $+9.3\text{pp}$), the sole exception being the Qwen2.5-3B 16K gate-misfire cell, and the partition claim rests on the unambiguous pooled NIAH-MK3 contrast (2/526 inputs recover). The full bootstrap protocol and per-cell interval discussion are deferred to Appendix A and Appendix G.

The Qwen2.5-3B 4K SnapKV cell decomposed along the budget axis (Table 8) and the 8-cell cross-(model, context) scaling table (Table 9, whose pattern matches the context-length scaling formula) are deferred to Appendix A.

7.2 A PRE-REGISTERED OUT-OF-RANGE TEST AT 32K

All scaling evidence so far is in-range (4K/16K), so we ran a pre-registered extrapolation test: fit the recovery ratio $\rho/(1 - A_{\text{full}})$ on 4K/16K, measure A_{full} at 32K (Qwen2.5-1.5B, $N = 100$), freeze and hash the predicted ρ intervals before sweeping (protocol log and SHA-256 released; table in Appendix A).

One of three predictions lands strictly inside its interval (Table 10). Both misses are in the more-recovery direction, but they differ in kind: VT recovering more than predicted is conservative for the scaling relation (the relation under-predicted the effect), whereas NIAH-MK3’s point estimate rising above the $\rho \leq 0.02$ pin (to 0.04) is a partition miss in the anti-conservative direction (the capacity-bound task recovered slightly more than the partition allows, though its Wilson interval still covers 0.02). A second input draw changes the picture in one place: FWE and NIAH-MK3 replicate (inside the band, near the pin), but VT is not seed-robust, sitting above the band on the first draw ($\rho = 0.14$) and inside it on the second ($\rho = 0.09$; 2 of 3 predictions inside on the second draw versus 1 of 3 on the first). We therefore do not claim the first draw’s VT amplification (that eviction beats full-KV on VT in mean accuracy). What survives both draws is the ordinal content, the partition is visible in the ratio at any context length (MK3’s ratio stays far below VT’s), while constant-ratio proportionality does not. We claim the formula as a directional and ordinal relation, not a quantitative one.

7.3 THE ACCURACY-MEMORY FRONTIER

What cache size does the gate actually deliver? Comparing at matched *achieved* kept-KV (Figure 2): gating improves the frontier by 3–19pp over plain SnapKV once the target keeps more than a precise crossover (≈ 0.32 – 0.35 of the cache), and below it cannot compete (its full-KV fallback floors compression at $\approx 3\times$). PAGE is a moderate-compression safety method, not an aggressive-regime compressor; against a per-input oracle it recovers a mean 69% of the headroom, almost all task-level protection of the capacity-bound anchor. Details and the exact crossover per model are in Appendix A.

Drop ordering and scaling-formula check. The head-agreement drop orders the tasks the same way in every Qwen and Mistral cell (NIAH-MK3 smallest; Figure 1 shows one cell), fully preserved on Qwen/Mistral and partially on the Llama family (Table 12, App. A.10). Verifying Eq. (4) on the 4K/16K cells, the ratio $\rho/(1 - A_{\text{full}})$ clusters in $[0.25, 0.55]$ on dilution-prone tasks and stays ≤ 0.02 on NIAH-MK3 regardless of headroom. Both the drop table and the scaling verification (Table 13, Figure 6) are in Appendix A.

7.4 SINGLE-TASK SHOWCASE: MISTRAL 4K NIAH-MK3

The strongest single-task win (Figure 3). $N = 100$. Plain SnapKV at $b = 0.0625$ collapses from 0.99 full-cache to 0.00; gated SnapKV holds 0.89 across all budgets. The gate is closed on 90/100 inputs (correct prediction of capacity-bound) with 10/100 false positives (gate opens, eviction destroys the needle).

7.5 COST OF THE GATE, AND WHAT THE DEPLOYMENT DELIVERS

The gate costs 17ms on Qwen2.5-1.5B and 89ms on Mistral-7B and yields a $1.29\times$ decode speedup at 7B; the full end-to-end efficiency table (Table 14), the per-layer ablation, and the random-eviction control are deferred to Appendix A.

7.6 HEAD-TO-HEADS WITH PUBLISHED METHODS

The gate wraps published evictors, not just the base four: around a CapKV re-implementation (Yang et al., 2026a) on LongBench it is weakly positive (+1.5/+8.3pp on two subtasks), and around pre-

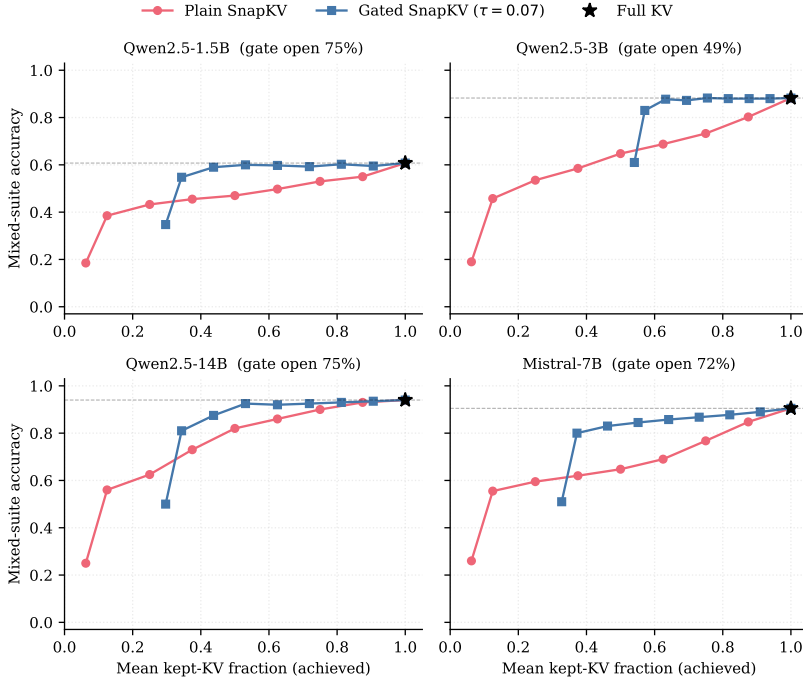


Figure 2: Accuracy vs. *achieved* kept-KV fraction (mean over inputs of n_{kept}/T) on the mixed 4K suite (NIAH-MK3 + VT + FWE + QA_1), SnapKV. Because the gate falls back to the full cache when it closes, gated points sit at larger kept-KV than their nominal budget; the honest comparison is at matched achieved cache size. For every target kept-KV fraction $\gtrsim 1/3$ (Qwen3B: $\gtrsim 0.57$) the gated curve is the Pareto frontier, by up to +19pp over plain at equal cache size; at the most aggressive nominal budget ($b = 0.0625$, leftmost gated point) the gated point is dominated by plain eviction at a matched cache size, and $\tau = 0.07$ caps achievable compression at $1 - \text{open-fraction}$ of the cache (0.25–0.51 across models). All gated results use the post-hoc $\tau = 0.07$ reconstruction (Appendix H); the reconstruction reproduces the stored gated outcomes exactly on the three runs executed at $\tau = 0.07$.

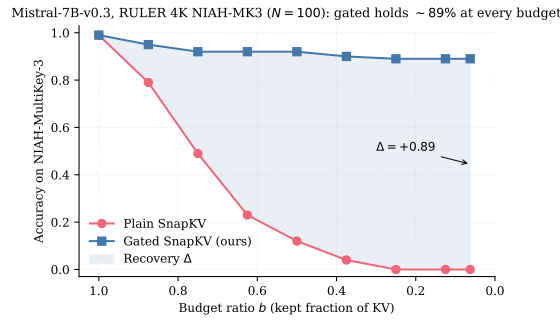


Figure 3: Mistral-7B 4K NIAH-MK3, $N = 100$. Accuracy vs. KV budget for plain SnapKV (solid, collapses to 0) and gated SnapKV (dashed, 0.89 flat across budgets). The gate prevents the collapse without retraining the base evictor or seeing labels.

trained DBTrimKV (Bui et al., 2026) (Qwen3-4B, RULER 4K) it improves matched-budget accuracy by +23.3pp. But the DBTrimKV number is a memory-spending win: the gate fires on 1.7% of records, holding $\approx 14.5\times$ the cache, and at matched *memory* ($\approx 30\%$ kept-KV) natively-run plain DBTrimKV *beats* the wrapped version. PAGE does not advance the frontier over a strong trained method at matched memory; its value is the label-free per-input decision to keep the full cache when eviction is unsafe (tables and the matched-memory head-to-head in Appendix A).

8 LIMITATIONS

We list the limitations of the present work in roughly decreasing order of severity.

The gate needs materialized prefill attention, which fused FlashAttention serving does not provide. D is computed from per-head prefill attention over the w observation queries. Production long-context serving (paged FlashAttention) never materializes the attention matrix, so the gate cannot be read off a stock serving stack: it requires a separate attention-exposing pass. Our two-pass mode does exactly this (SDPA/flash prefill plus an eager re-forward of the last w queries against the cache), keeping the extra pass to $O(w \cdot T)$ rather than $O(T^2)$, but it is still an additional pass a vanilla stack does not run, and its $O(LH^2k)$ cost grows with head count (Section 7.5). PAGE is therefore best suited to deployments that can afford one attention-exposing prefill pass (or that already run one for attention-scored eviction); it is not a drop-in for a fused-kernel serving path, and its memory benefit is bounded by $1 - p_{\text{open}}$.

The theory is a conditional analysis on a surrogate. On a single-head deterministic-SNR surrogate we give (Appendix I) a two-sided bound on the scaling relation, a converse “capacity-bound $\Rightarrow$ small D ” that is only non-vacuous at a large margin (so it constrains the drop *ordering* more than its sign), and an approximate (not exact) useful-set identification (A4). These are conditional results on a toy model; the paper’s claims rest on the empirical partition and the orthogonality demonstration, neither of which uses the bridge.

The fixed τ does not transfer to the Llama family, and requires a per-model pilot. On Yi-1.5-9B and Llama-3.1-8B the ordering transfers but the fixed $\tau = 0.07$ does not: it mis-classifies `niah_multivalue` on both, and on Yi the drop-vs- ρ prediction is inverted. The z-scored variant (Section 5) repairs this with an unlabeled per-model pilot (Llama Δ to +0.174, Yi compressing at +0.145), but that is a per-model calibration step, its task-level claim excludes `niah_multivalue`, and per-input separation stays at $\text{AUC} \approx 0.80$; a per-input statistic beyond mean D is the natural follow-up (Appendix A).

NIAH-MK3 is capacity-bound scorer-independently, but only at aggressive compression. To rule out a scorer-specific effect we re-implemented ManifoldKV (Datta et al., 2026) faithfully, with Ada-KV per-head budget allocation (Feng et al., 2025) realized exactly by per-head masking. The scorer is demonstrably competent: it reproduces ManifoldKV’s signature multi-key advantage on *two*-key NIAH, holding 0.72 accuracy at 50% cache where SnapKV collapses to 0.16. Yet the same scorer does not rescue three-key NIAH-MK3: below 25% budget it falls to chance like every attention-score evictor, and never exceeds a per-head SnapKV baseline (Appendix A). NIAH-MK3 is thus capacity-bound independently of the scoring family. Honest caveat on magnitude: at $2\times$ a clean per-head SnapKV drops MK3 only to ≈ 0.32 (from 0.66), whereas our default single-mask evictor reports a sharper 0.14, so the moderate-budget collapse in our matrix and showcase is partly a single-mask artifact; the scorer-independent absolute collapse is at $\geq 4\times$, where the gate’s protection matters and where the showcase should be read.

Further limitations, all in Appendix A: greedy runs with low-powered per-cell partition calls (a second draw replicates transfer and head-to-heads but not the 32K VT amplification); a near-zero gate effect on the realistic LongBench subset and on multi-hop hotpotqa (both evict-robust and correctly gated open, but neither a capacity-bound anchor, which remains RULER-only), with one negative cell (-2.2pp); a false-positive gate call on `passage_count`; and head-to-heads run on different models.

9 CONCLUSION

The benefit of KV-cache eviction is sharply partitioned across inputs, and one label-free prefill statistic, the early-to-late head-agreement drop, predicts which side an input falls on. Read honestly, PAGE is a safety mechanism: it lifts the accuracy–memory frontier where moderate compression is the target and prevents catastrophic collapse where it is not (Mistral-7B NIAH-MK3, $99\% \rightarrow 0\%$ becomes a flat 89%), at the cost of one attention-exposing prefill pass. The open problems are a

signal that transfers across families without a per-model pilot, and a realistic non-synthetic capacity-bound benchmark.

REPRODUCIBILITY STATEMENT

Our SnapKV-family runs (the partition, the matrix, the scaling and 32K sweeps, and the LongBench probes) use single NVIDIA A100 80GB GPUs with PyTorch 2.11, Transformers 5.9, and Python 3.13, in bfloat16 with greedy (argmax) decoding, `max_new_tokens = 128`, and a fixed seed. The two published head-to-heads run in the baselines’ own pinned environments (the DBTrimKV comparison uses the authors’ `trimkv` environment, Transformers 4.57.1 / PyTorch 2.8 / FlashAttention 2; the CapKV comparison is our re-implementation in an isolated environment), documented with the released logs. Models are the public Hugging Face checkpoints of Qwen2.5-1.5B/3B/14B-Instruct, Mistral-7B-Instruct-v0.3, Yi-1.5-9B-Chat, and Llama-3.1-8B-Instruct; tasks come from the `simonjegou/ruler` port of RULER and from LongBench. Cache slicing preserves RoPE alignment via explicit `position_ids`; 16K runs use SDPA prefill with an eager re-forward of the $w = 32$ observation window for scoring (Appendix H). Per-cell sample sizes are stated in each table caption. The gate is one thresholded scalar (Eq. (3)), so every gated number in the paper can be recomputed from the released per-input logs; code, sweep configurations, and raw per-input jsonl outputs will be released with the camera-ready.

ETHICS STATEMENT

This work studies inference-time efficiency of publicly released language models on public benchmarks. It involves no human subjects, no personal data, and no new model training. Better KV-cache eviction lowers the memory and energy cost of long-context inference; we see no ethical risk specific to this method beyond those of the underlying models.

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A DEFERRED TABLES, DERIVATIONS, AND SECONDARY EXPERIMENTS

A.1 RELATED WORK (FULL TREATMENT)

Five close concurrent neighbors. Bui et al. (2026) (DBTrimKV) is the most important reference point. Their Proposition 3.1 establishes that near-tie distractors force attention dilution in a single head, and Corollary 3.2 shows preferential retention of high-attention positions reduces it; Section 6 adopts this mechanism. Their evaluation reports DBTrimKV > full-cache on multi-turn dialogue but does not characterise a task family on which their method underperforms full-cache. They train a retention gate end-to-end; our gate is label-free and wraps theirs (and any other) without retraining.

Direct head-to-head: Section 7.6, $\Delta = +0.233$ at matched budget on RULER 4K Qwen3-4B-Instruct.

Yang et al. (2026a) (CapKV) derives a closed-form, label-free eviction score from an information-bottleneck objective and reports gains over a Vanilla baseline on LongBench. Their score is task-agnostic; we partition tasks into a class for which any eviction is risky and a class for which it helps, and report a direct head-to-head wrapping CapKV in our gate (Section 7.6, $\Delta = +1.5$ to $+8.3$ pp at matched budget on two LongBench subtasks).

Zhou et al. (2025) (DynamicKV) introduces task-aware per-layer adaptive budgets and explicitly observes that the layer-wise attention pattern differs across task families, using attention statistics to drive a continuous per-task budget allocation. This is the closest conceptual neighbor to our partition: both detect that task type modulates which KV positions matter. The differences are (i) DynamicKV’s per-task budget is fitted per task with calibration data; ours is a per-input gate triggered by a single observable statistic with one fixed threshold; (ii) their benchmark suite (LongBench) does not include a task on which eviction strictly damages accuracy, so the binary partition is not visible in their results; (iii) DynamicKV does not formalise a partition predictor or attempt cross-architecture transfer. We therefore see our partition + drop predictor + binary gate as orthogonal but partially overlapping with their empirical observation, and we acknowledge the overlap.

Garcia (2026) (“Protection Is All You Need”) shows that structural protection (sink tokens plus a recent-token window) dominates the choice of scorer on globally capacity-capped harnesses. Our experimental setup uses protection of the same kind but smaller extent ($n_{\text{sink}} = 4$ sink tokens and a $w = 32$ observation window, versus the 10% prefix plus 10% suffix they advocate). A protection-matched ablation at their 10% + 10% setting confirms the partition is not an artifact of light protection: NIAH-MK3 still collapses and the eviction-helps signal survives (Section 3).

Yang et al. (2026c) (IndexMem) trains a learnable indexer plus latent memory and observes an information-density effect on LongBench averages around 50% compression. The benchmark-average hides which sub-tasks contribute the peak; our partition makes that explicit, and our predictor is training-free.

Concurrent 2026 diagnostics and wrappers. Zhang et al. (2026) ask a similar headline question (when does value-aware KV eviction help?). The clearest separation is the axis: their score is *value-aware* (a per-input support-coupling score $\phi(x)$ over value states), whereas our signal is *attention topology*, a complementary account in the same sense that VaSE is (below). Their unit of analysis is the selector rather than the task, their $\phi(x)$ drives diagnostic grid splits rather than an evict-or-not gate, and they report that diagnostic signals transfer across selectors only conditionally, which makes the single transferred threshold we report non-obvious. Chen et al. (2026) document instruction-type-dependent degradation under five eviction methods (some instruction types become entirely ignored), the closest published task-level failure taxonomy; we cede the hurts-degradation observation to them and claim the joint helps-versus-hurts partition plus the a-priori predictor. Ananthanarayanan et al. (2026) compute a *within-layer* head-consensus level (how much heads inside a layer attend to the same tokens) and relate it descriptively to compression tolerance at the architecture level. Our statistic is different: an *early-to-late*, *cross-layer* drop in the top- k agreement between heads, computed online at prefill and thresholded per input to gate. To our knowledge the identical statistic has not been proposed as an eviction gate; the ablation of Appendix A shows it separates the partition cleanly (AUC 1.000) and transfers across families. On the failure side, Chang et al. (2026) (VaSE) trace catastrophic eviction failures to high-magnitude value states, an axis orthogonal to our attention-topology account, and ManifoldKV (Datta et al., 2026) shows that a Euclidean-outlier key scorer improves NIAH-MultiKey retention under compression, which is why we define “capacity-bound” relative to the scorer being gated (Section 8). On the wrapper side, VECTOR (Lin et al., 2026b) and CriticalKV (Feng et al., 2026) are plug-and-play augmentations over multiple base evictors; both always compress under a fixed budget and adapt *which* tokens are kept, whereas our gate decides *whether* to evict at all. Cross-family transfer of an eviction control is itself known, via CompilerKV’s (Yang et al., 2026b) portable per-head retention tables; we transfer one threshold instead, fixed across the Qwen and Mistral families and extended to the Llama family and Qwen3 by an unlabeled per-model standardization. Further concurrent evictors and allocators, EntropyInfer (Xu et al., 2026), CompressKV (Lin et al., 2026a), MomentKV (Li et al., 2026b), EpiKV (Anonymous, 2026), DefensiveKV (Anonymous, 2025a), and TAKE (Anonymous, 2025b),

adapt budgets or scores per head, layer, token, or task, but none gates the eviction decision per input or identifies a task class where eviction hurts monotonically. Learned gates (Attention-Gate (Anonymous, 2024), Fast KVzip (Kim et al., 2026)) share the “gate” name but train their gating parameters, unlike our inference-time scalar.

Older anchors. SnapKV (Li et al., 2024), H2O (Zhang et al., 2023), StreamingLLM (Xiao et al., 2024), PyramidKV (Cai et al., 2025), and Ada-KV (Feng et al., 2025) are the standard training-free eviction baselines. We use SnapKV-style scoring as the base for our gate and demonstrate orthogonality on H2O, StreamingLLM, and PyramidKV. ForesightKV (Dong et al., 2026) and Learning-to-Evict (Moschella et al., 2026) train predictors of future utility; they require labels or rollouts. UT-ACA (Zhou et al., 2026) and LU-KV (Tang et al., 2026) are calibration-based context allocators on a single substrate. None proposes a binary task-type partition or an a priori predictor.

Adjacent. AdaCompute-GBM (Zhai et al., 2026) adapts on the self-consistency sample-count axis; BudgetThinker (Wen et al., 2025) on the CoT chain-length axis; LaTER (Li et al., 2026a) on latent-then-explicit reasoning. These work on substrates orthogonal to the KV-cache axis we study.

A.2 ADDITIONAL EMPIRICAL-PARTITION ANALYSES

Qwen2.5-14B: re-entry to the dilution regime at long context. At 4K, three dilution-prone tasks (VT, FWE, QA_2) sit at $\rho \approx 0$: saturation, since $A_{\text{full}} \in [0.92, 1.00]$ leaves no headroom for recovery, as Eq. (4) predicts. At 16K, two of five dilution-prone tasks cross the threshold (FWE $0.00 \rightarrow 0.20$, QA_1 $0.06 \rightarrow 0.10$), niah_multivalue stays above it ($0.08 \rightarrow 0.07$), and the capacity-bound NIAH-MK3 stays at exactly 0.00 at both scales. VT ($0.00 \rightarrow 0.03$) and QA_2 ($0.02 \rightarrow 0.03$) remain below threshold because the inputs that the model gets wrong at 4K stay wrong at every budget at 16K. Headroom opens, dilution returns, NIAH-MK3 stays pinned: the partition behaves as the scaling formula dictates, not as task identity dictates.

The partition is not a task-name heuristic. The NIAH family contains structurally different sub-tasks: niah_multivalue (one key, many values) is dilution-prone in seven of the eight cells measured (Table 1), while NIAH-MK3 (three near-tie keys, one of which is the answer) is capacity-bound in all of them, and the distractor sweep below places niah_multikey_1 (no near-tie distractors) on the dilution-prone side as well. The *near-tie distractor* structure, not the surface task category, is what matters, and the head-agreement drop in Section 4.1 catches the distinction. A non-RULER probe confirms this: LongBench passage_retrieval_en (pick 1 of 30 paragraphs; $N = 100$, Qwen2.5-1.5B, prompts 10–16K tokens) *looks* NIAH-like but has no near-tie distractors, and it is evict-robust in practice, accuracy stays flat from $b = 1.0$ to $b = 0.0625$ ($0.29 \rightarrow 0.26$, $\rho = 0.01$), while its mean drop $D = 0.105$ sits squarely in the dilution-prone cluster, so the gate opens on 100/100 inputs. The predictor gets the surface-similar, structurally different task right a priori.

The partition survives Garcia-scale protection. Garcia (2026) argue that structural protection dominates scorer choice, so we re-ran the Qwen2.5-1.5B 4K partition tasks with their 10% prefix + 10% suffix protection ($n_{\text{sink}} = w = 410$; $N = 100$ per task). One accounting note: protection overrides the budget, so all nominal $b \leq 0.1875$ share an effective floor of 824 kept tokens (≈ 0.21 of context). Under this heavy protection the partition is unchanged. NIAH-MK3 still collapses (0.65 full-KV $\rightarrow 0.09$ at the floor, an 86% relative loss despite the floor cache being $3.3\times$ our light-protection cache at nominal $b = 0.0625$), so the capacity-bound failure is not protection-fixable. The eviction-helps signal survives on the dilution-prone side (ρ : FWE $0.06 \rightarrow 0.13$, QA_1 $0.08 \rightarrow 0.07$, VT $0.06 \rightarrow 0.03$; aggregate $0.077 \geq 0.05$), and FWE’s floor accuracy (0.31) exceeds its full-KV accuracy (0.22). Heavy protection is also not free: protected VT drops to 0.00 at the floor while our light-protection run holds 0.75 at nominal $b = 0.125$ protected prefix/suffix tokens crowd out the scorer-selected middle tokens the task needs. Caveat: in our harness w is both the recency protection and the SnapKV scoring window, so this run is a protection-heavy SnapKV variant rather than an exact replication of their protocol.

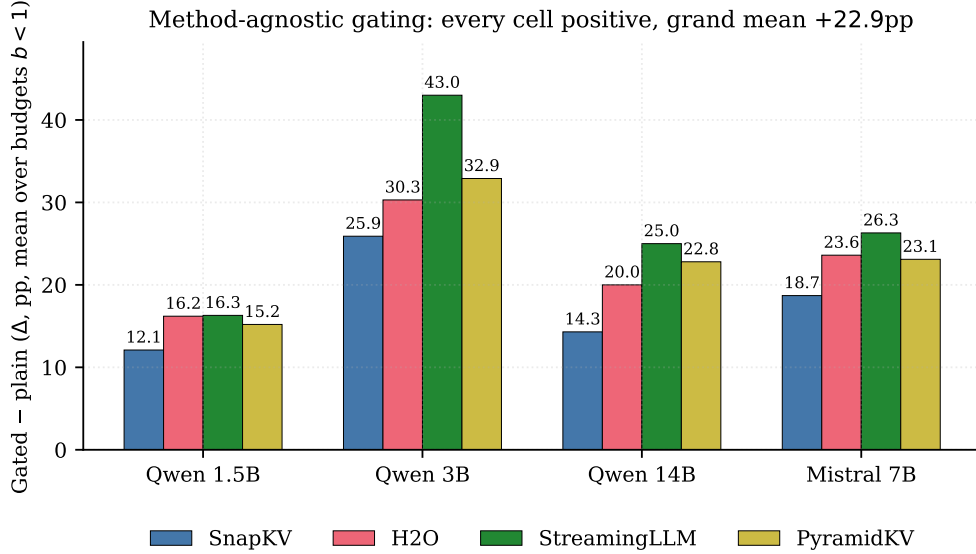


Figure 4: The headline 4×4 method-agnostic matrix. Bars are gated – plain accuracy averaged over budgets. The wrapper is positive in every one of 16 cells with a single $\tau = 0.07$ and no per-base-evictor tuning. Grand mean +22.9pp.

A.3 HEADLINE MATRIX FIGURE AND A GEOMETRY-BASED BASE EVICTOR

A fifth, geometry-based base evictor. To test composition beyond attention-score-based bases, we add a ManifoldKV-style scorer (Datta et al., 2026): per-(layer, head) Euclidean distance of each key to the head’s key centroid, averaged into a single shared keep mask (a single-mask variant of their per-layer design, forced by our uniform-cache harness). On the Qwen2.5-1.5B 4K mixed suite ($N = 100/\text{task}$, four budgets), the same gate at the same $\tau = 0.07$ lifts it by $\Delta = +16.2\text{pp}$ with no negative (task, budget) cell; on NIAH-MK3 the gate recovers full-KV accuracy ($0.00 \rightarrow 0.65$) at every budget. Two observations: the gate signal is attention-based and scorer-independent, so composition extends outside the attention-score family; and our single-mask variant does *not* reproduce ManifoldKV’s reported multi-key rescue (plain accuracy 0.01 at $b = 0.5$), their per-layer, per-head design may behave differently, so we treat their published numbers as the authoritative claim and scope ours accordingly (Section 8).

A.4 ADDITIONAL LIMITATIONS

Single-seed runs; some cells are low-powered. All sweeps use greedy decoding and a single fixed seed; per-cell N ranges from 30 to 100 inputs (per-task N noted in each table). All headline deltas carry cluster-bootstrap 95% intervals and all ρ values Wilson intervals (Section 7, Appendix G); every matrix delta excludes zero, but the per-cell partition calls are individually low-powered (most straddle the 0.05 threshold), and small- N cells (Qwen-14B 16K, $N = 30$; triviaqa head-to-head, $N = 24$) remain noisy. The partition claim rests on the pooled contrast, which is unambiguous.

LongBench-SnapKV shows $\Delta \approx 0$ on the dilution-prone subset. On LongBench (Bai et al., 2024) qasper, triviaqa, trec, and multifieldqa_en under SnapKV scoring (plus passage_retrieval_en on the 1.5B model), the pooled gated Δ is within $\pm 1\text{pp}$ of plain on all four models (Qwen2.5-1.5B 0.000, Qwen2.5-3B +0.002, Mistral-7B +0.003, Qwen2.5-14B -0.008). The worst per-task cell is Qwen2.5-14B trec at -2.2pp, the only negative gating result in the project, which we attribute to gate misfires on a task with no capacity-bound inputs to rescue. LongBench, on this subset, has no capacity-bound task; the partition predicts neutrality and we observe it, including its mildly negative tail.

Table 3: Distractor-count sweep on the NIAH-MultiKey family (Qwen2.5-1.5B, RULER 4K, $N = 50$ per task). Mean head-agreement drop D shrinks monotonically as the near-tie distractor count grows.

Task	mean D	gate-open frac	ρ_{plain}	ρ_{gated}	plain at $b=0.0625$
niah_multikey_1	+0.155	1.00	0.04	0.04	0.36
niah_multikey_2	+0.122	1.00	0.00	0.00	0.00
niah_multikey_3	+0.045	0.00	0.00	0.00	0.00

Qwen2.5-3B 16K needs τ -calibration. The fixed $\tau = 0.07$ default fails on this cell. The 40-input recipe of Section 5 fixes it (Δ goes from +3.4pp to +5.4pp). We recommend the recipe as the deployment default; we keep the fixed- τ result throughout to show the gap is small.

DBTrimKV head-to-head uses Qwen3 not Qwen2.5. DBTrimKV’s pretrained gates exist only for Qwen3 variants, so Section 7.6 runs on Qwen3-4B-Instruct-2507. The matrix is Qwen2.5-keyed, so DBTrimKV stays a standalone head-to-head rather than a matrix row.

4 base evictors, not 6. The matrix covers training-free policies. Trained / learned evictors (DB-TrimKV, CapKV, IndexMem, DynamicKV) are cited but not in the matrix for the codebase-compat reason above.

One confirmed capacity-bound exemplar, and a documented false positive. NIAH-MK3 is the only task where empirical capacity-boundedness (accuracy collapse under eviction) and the gate’s capacity-bound classification coincide. On LongBench passage_count at Qwen2.5-14B the gate closes on all inputs (mean $D = -0.013$) yet eviction is empirically *safe* (accuracy flat under $16\times$ eviction, $\rho = 0$): a benign *false positive* (over-conservative, costing compression not accuracy), the dual of the MK2 false negative (Appendix A). So the head-agreement drop is a one-sided predictor, reliable at firing on dilution-prone inputs, not a perfect classifier. Multilingual and code tasks are out of scope; we expect the mechanism to transfer because it is architecture-level, but we have not measured it.

A.5 MECHANISM-VALIDATION DETAIL: DISTRACTOR COUNT DRIVES D

The RULER NIAH-MultiKey family lets us sweep the near-tie distractor count directly: niah_multikey_1 (no distractors), niah_multikey_2 (one near-tie distractor), niah_multikey_3 (two near-tie distractors). The dilution mechanism of Bui et al. (2026) predicts that more near-tie distractors should produce larger expected dilution δ_t and (via App. F) smaller head-agreement drop D . On Qwen2.5-1.5B at RULER 4K, $N = 50$ per task, we observe a clean monotone gradient (Table 3).

D shrinks by $\sim 3.5\times$ from $0 \rightarrow 2$ near-tie distractors, and per-input ranges between MK2 and MK3 do not overlap ($\max_{\text{MK3}} D = 0.062 < \min_{\text{MK2}} D = 0.088$): the partition predictor separates 2-distractor inputs from ≤ 1 -distractor inputs 100% of the time at $\tau = 0.07$. The monotone gradient replicates on Mistral-7B-v0.3 ($D = 0.081, 0.072, 0.060$ for MK1/MK2/MK3, $N = 50$ per task), confirming the mechanism is not Qwen-specific, though the finer per-input MK2/MK3 non-overlap is (their ranges overlap on Mistral). On a controlled sweep over distractor count, the head-agreement drop and the partition track each other monotonically, exactly as the δ_t dilution mechanism predicts.

The single-near-tie boundary case. MK2 is empirically capacity-bound at low budgets (plain SnapKV collapses from 80% at $b=1$ to 0% at $b=0.0625$) yet is classified as dilution-prone by the predictor at $\tau = 0.07$: MK2’s mean $D = 0.122$ sits between the MK3 cluster ($D \leq 0.062$) and the MK1 cluster ($D \geq 0.099$), co-located with genuinely dilution-prone FWE ($D = 0.098$) and niah_multivalue ($D = 0.107$). No single mean- D threshold separates MK2 from the dilution-prone cluster on this model. The predictor therefore falls back to its weakest form: it correctly classifies MK3 and the standard dilution-prone tasks but mis-classifies single-near-tie MK2. Per-input recovery information beyond mean D would be needed to catch it at deployment.

Table 4: Gate-signal ablation on Qwen2.5-1.5B-Instruct (RULER 4K, $N = 50$ per task, one shared prefill per input). Each cheap prefill statistic is tested for separating the capacity-bound anchor (NIAH-MK3, and MK3+MK2) from the dilution-prone tasks (vt, fwe, qa_1, multivalue). “Sep.” = a single threshold puts the capacity-bound task mean(s) strictly on one side of all four dilution task means; AUCs are per-input and oriented per signal (0.5 = chance); “MK mono.” = task means are monotone along the MK1→MK2→MK3 distractor gradient; r = Pearson correlation with D .

Signal	Sep. MK3	Sep. MK3+MK2	AUC MK3	AUC MK3+MK2	MK mono.	r w/ D
head-agreement drop D (paper)	✓	–	1.000	0.859	✓	+1.00
mean attention entropy (norm.)	–	–	0.508	0.670	–	+0.08
top-32 attention mass share	–	–	0.784	0.574	✓	+0.22
max single-position share	–	–	0.723	0.708	✓	+0.32
key-norm dispersion (std/mean)	–	–	0.570	0.560	–	+0.01
early–late entropy drop	–	–	0.805	0.575	–	–0.64

Table 5: Cross-architecture SnapKV-only runs on two Llama-family models. These use the four most aggressive budgets $\{0.0625, 0.125, 0.25, 0.5\}$ rather than the eight-budget grid of Table 2, so the Δ values are not directly comparable to the matrix rows (aggressive-budget-only averaging inflates Δ).

Model	plain SnapKV	gated SnapKV	Δ
Yi-1.5-9B	0.464	0.854	+0.390
Llama-3.1-8B	0.509	0.618	+0.109

A.6 GATE-SIGNAL ABLATION: WHY THE HEAD-AGREEMENT DROP

A same-inputs ablation against five other cheap prefill statistics (Table 4; one shared prefill per input, $N = 50$ per task across seven RULER tasks) shows D is not merely the best candidate but the only one that separates the partition at all: MK3’s mean D sits strictly below all four dilution-prone task means with per-input AUC 1.000, while every alternative fails task-level separation and tops out at AUC 0.805. The dissection is informative: raw attention entropy is at chance (0.508), taking its early-vs-late layer contrast lifts it to 0.805, and replacing entropy with head *agreement* makes the separation exact, both the layer contrast and the agreement measure matter, with agreement decisive. No alternative is even a redundant proxy (all $|r|$ with $D \leq 0.64$). Consistent with the MK2 boundary case of Section 3, extending the capacity-bound anchor to MK3+MK2 breaks task-level separation for D too (AUC 0.859): D tracks the near-tie distractor gradient, not a binary task label.

A.7 CROSS-ARCHITECTURE, SCALE, AND BEST-PLAIN RESULTS

Yi-1.5-9B and Llama-3.1-8B cross-architecture runs (Table 5). Yi ($\Delta = +39.0\text{pp}$) and Llama ($\Delta = +10.9\text{pp}$) are Llama-architecture descendants distinct from Qwen and Mistral; only the SnapKV column was run for compute, and on a smaller, more aggressive budget grid (see caption). The sign is positive on both, but on these two models part of the gain comes from the gate closing conservatively on tasks where plain SnapKV collapses, rather than from a correctly recovered partition. The per-task breakdown and the shared Llama-arch family caveat (niah_multivalue mis-classifies on both Yi and Llama) are in Section A.10 and Section 8.

Scale validation at 32B. A SnapKV-only run on Qwen2.5-32B-Instruct (RULER 4K mixed suite, $N = 100$) confirms the predictor scales but also illustrates the saturation arm of the scaling formula. The drop ordering transfers cleanly (NIAH-MK3 smallest at $D = 0.025$, then FWE 0.052, VT 0.079, QA_1 0.103). At the same time the 32B model is near-saturated at 4K ($A_{\text{full}} = 1.00$ on VT and NIAH-MK3, 0.93–0.94 on FWE/QA_1), so by Eq. (4) there is essentially no dilution headroom and the gate’s value is capacity-bound protection rather than recovery: plain SnapKV collapses at $b = 0.0625$ (NIAH-MK3 0.00, VT 0.00, FWE 0.09) while the gate holds the full cache on the low-drop tasks, giving mean $\Delta = +0.335$ at $\tau = 0.07$ (driven by NIAH-MK3 +0.88 and FWE +0.44) and +0.237 at the z-scored threshold with more compression (0.45 vs. 0.63 kept-KV). We report this as a scale point for the partition and predictor, not as additional dilution evidence.

Table 6: Best plain evictor vs. best gated evictor per model, over the four base evictors of Table 2 at the same budget grid.

Model	Best plain	Best gated	Δ
Qwen2.5-1.5B	0.438 (SnapKV)	0.570 (PyramidKV)	+0.132
Qwen2.5-3B	0.580 (SnapKV)	0.865 (PyramidKV)	+0.285
Qwen2.5-14B	0.709 (SnapKV)	0.868 (H2O)	+0.159
Mistral-7B	0.623 (SnapKV)	0.823 (PyramidKV)	+0.201

Table 7: Calibration-recipe validation. $\bar{D}_C$ uses NIAH-MK3 ($N = 20$); $\bar{D}_D$ uses VT ($N = 20$). Bolded τ -choice column is the winner per cell. The pilot means here are computed on 20-input samples and therefore differ slightly from the $N \geq 50$ per-task means of Table 12.

Cell	$\bar{D}_C$	$\bar{D}_D$	τ_{cal}	gated@ τ_{cal}	gated@ $\tau=0.07$
Qwen2.5-1.5B 4K	+0.045	+0.161	+0.103	+0.121	+0.121
Qwen2.5-3B 4K	-0.002	+0.076	+0.037	+0.213	+0.259
Qwen2.5-3B 16K	-0.021	+0.068	+0.024	+0.054	+0.034
Mistral 4K	+0.060	+0.096	+0.078	+0.223	+0.187

Best plain vs. best gated. A natural concern is “you only beat the weakest baseline.” Comparing the best plain evictor against the best gated evictor for each model rules this out (Table 6).

Why the deltas vary across base evictors. StreamingLLM keeps only 36 tokens regardless of budget; on the dilution-prone fraction it is near zero plain, near zero gated, so gating contributes only via the gate-closed NIAH-MK3 inputs that go from 0 to full-KV accuracy. SnapKV is budget-scaled and degrades more gracefully, so it has less room for gating to recover. PyramidKV’s depth-weighted scoring biases toward bottom layers but produces similar absolute gains to SnapKV. The pattern is that gating Δ tracks how much the base evictor collapses on the capacity-bound fraction. Two implementation caveats: our PyramidKV is a depth-weighted variant of the published per-layer budget schedule, and under two-pass prefill scoring (used on Qwen2.5-14B for memory) the H2O score only sees the last w queries, so the 14B H2O column degenerates toward SnapKV-style scoring; its Δ there should be read as a near-duplicate of the SnapKV cell rather than an independent base evictor.

A.8 CALIBRATION RECIPE DETAILS

We provide a one-shot, label-free calibration recipe.

Recipe. For a new (model, context):

1. Pick 20 inputs from a known capacity-bound task (NIAH-MK3) and compute the head-agreement drop D_i on each. Average to get $\bar{D}_C$.
2. Pick 20 inputs from a known dilution-prone task (VT) and compute $\bar{D}_D$.
3. Set

$$\tau = \frac{1}{2}(\bar{D}_C + \bar{D}_D). \quad (5)$$

40 prefills, no labels, no decode, < 5 minutes on a single A100.

Validation. On four cells (Table 7). The recipe lifts Qwen2.5-3B 16K from +3.4pp to +5.4pp (the only failure cell turned into a win) and lifts Mistral 4K from +18.7pp to +22.3pp. It is neutral on average across the four cells but with smaller variance.

τ sensitivity. Sweeping $\tau \in \{0.01, 0.025, 0.04, 0.055, 0.07, 0.085, 0.10, 0.13, 0.16, 0.20\}$ on Qwen2.5-1.5B at RULER 4K (mixed suite, $N = 100$, budgets $\{0.5, 0.25, 0.125, 0.0625\}$, SnapKV)

shows an operational plateau $\tau \in [0.055, 0.10]$ inside which Δ varies by under 1pp and both false-positive (gate fires on capacity-bound MK3) and false-negative (gate stays closed on dilution-prone VT/FWE/QA_1) rates stay at or below 9%. The deployed $\tau = 0.07$ sits in the middle of this plateau (Figure 5); outside it the behavior changes character (below 0.04: $\text{FP} \geq 64\%$, $\Delta \leq +0.06$; above 0.13: $\text{FN} \geq 33\%$ and the method degenerates toward “always full KV,” which on this full-KV-strong suite still yields a large Δ the sweep’s raw maximum is in fact at $\tau = 0.20$, but at the price of never compressing, so it is not a useful operating point). The deployed threshold does not depend on a knife-edge.

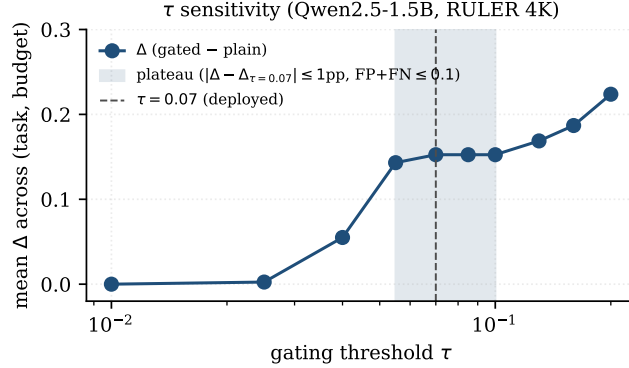


Figure 5: τ sensitivity on Qwen2.5-1.5B, RULER 4K. Mean Δ (gated – plain) over (task, budget) cells. Shaded band: operational plateau where $|\Delta - \Delta_{\tau=0.07}| \leq 1\text{pp}$ and $\text{FP} + \text{FN} \leq 0.10$. The deployed $\tau = 0.07$ sits in the middle of the plateau.

Architecture-normalized variant: the z-scored drop. The recipe above calibrates a per-cell threshold from two labeled pilot tasks. A stronger variant needs no task labels at all: standardize the drop per model, $z(x) = (D(x) - \mu_M)/\sigma_M$, with μ_M, σ_M pooled over an unlabeled ~ 100 -input pilot, and use one global threshold θ_z . Fitting $\theta_z = -0.69$ on the five Qwen/Mistral cells *only* and transferring it zero-shot to the two held-out Llama-family models: task-level separation of NIAH-MK3 from {QA_1, QA_2, VT, FWE} succeeds in 7 of 7 cells (pooled cross-cell AUC 0.965), where raw D with one threshold manages 4 of 7. The reason is that raw D is an architecture-invariant *ordering* whose location and scale are model-specific: $\tau = 0.07$ sits at $z = +0.69$ on Yi but $z = -0.97$ on Llama, two opposite failure modes that standardization removes. Downstream, the z-gate lifts Llama-3.1-8B from $\Delta = +0.109$ to $+0.174$ (NIAH-MK3 gate-open $0.56 \rightarrow 0.30$, MK3 accuracy $0.46 \rightarrow 0.72$) and gives Yi $\Delta = +0.145$ with *real* compression (kept-KV $0.22\text{--}0.32$ versus $0.76\text{--}0.89$ under the fixed- τ fallback). Two honest scopes: `niah_multivalue` must be excluded from the task-level ordering claim (its mean D sits below MK3 on both Llama-family models, which is behaviorally the right call there, since it is eviction-fragile on both), and per-input MK3-vs-rest AUC on the Llama family is intrinsically ≈ 0.80 ; no monotone rescaling changes within-model ranking. Shape-based alternatives (depth-correlation of per-layer agreement) fail outright (AUC 0.22 on Llama), confirming the raw drop is the right base statistic.

A fourth family where fixed τ fails outright: Qwen3. The starkest transfer test is Qwen3-4B-Instruct, a different model generation from the Qwen2.5 calibration set, whose drop distribution is compressed to a different scale entirely ($\mu = 0.034$, $\sigma = 0.017$ pooled over 400 inputs). There the fixed $\tau = 0.07$ sits at $z = +2.0$ and the gate essentially never fires: it keeps 99.4% of the cache and its apparent $\Delta = +0.43$ is a pure full-KV-fallback artifact with 0.6% compression. The z-scored threshold ($\tau_{\text{Qwen3}} = \mu + \theta_z \sigma = 0.022$, same $\theta_z = -0.69$) restores a working gate: NIAH-MK3 stays closed (open fraction 0.02, accuracy $0.05 \rightarrow 0.98$) while the three dilution-prone tasks open ($0.94\text{--}1.00$), yielding $2.3\times$ compression (43% mean kept-KV) at $\Delta = +0.235$ over plain and recovering 54.5% of the per-input oracle headroom. The NIAH-MK3-smallest ordering transfers cleanly. Qwen3 is thus the cleanest evidence for the paper’s central claim about the predictor: the drop is an architecture-invariant *ordering* whose scale must be standardized per model, here fixed- τ does not merely underperform, it does nothing.

Status. The recipe is the deployment-friendly default: it always fixes the failure cell, and the z-scored variant extends it label-free across architecture families. We report fixed- τ numbers throughout as the conservative default so readers see both the robustness of one constant and what standardization buys.

A.9 THEORY: MECHANISM, PARTITION, AND SCALING DERIVATIONS

Mechanism (adopted from Bui et al.). We adopt verbatim the dilution mechanism from Bui et al. (2026), which suffices to motivate our gating predictor. Fix a frozen decoder language model M . A single attention head reads from positions $t \in \{1, \dots, T\}$, each carrying a value v_t and a pre-eviction probability α_t with $\sum_t \alpha_t = 1$. Partition positions into a relevant set $\mathcal{R} = \{t : r_t = 1\}$ ($|\mathcal{R}| = R$) and a noise set $\mathcal{N}$ ($|\mathcal{N}| = N$), with $R \ll N$. Following Bui et al. (2026) Eq. (7), let $U_t \subseteq \mathcal{R}$ denote the *useful* set at step t (the positions whose retention preserves the next-token margin), and define dilution as the attention mass *outside* U_t :

$$\delta_t := 1 - \sum_{i \in U_t} \alpha_{t,i}. \quad (6)$$

(Notation rearranged to fit our manuscript; this is exactly their quantity.) Their Proposition 3.1 establishes that near-tie distractors (α_t comparable across $\mathcal{N}$) force δ_t to be large. Their Corollary 3.2 shows that preferential retention with $\rho_U \geq \rho_D$ (where ρ_U, ρ_D are the retained mass on useful and distractor positions) reduces δ_t . We refer to their paper for the full proof; we use this mechanism as given.

The partition follows from the mechanism. Define $\alpha_{\mathcal{R}} := \sum_{t \in \mathcal{R}} \alpha_t$ as the pre-eviction signal mass.

- *Dilution-prone (D):* $\alpha_{\mathcal{R}} \ll 1$. Most attention is on noise; eviction of noise positions monotonically improves the head SNR.
- *Capacity-bound (C):* $\alpha_{\mathcal{R}} \rightarrow 1$. Attention is concentrated on the few relevant positions. Imperfect scoring can only *lose* signal; eviction strictly degrades the head.

The partition we observe on RULER is the empirical projection of $\alpha_{\mathcal{R}}$ onto the task structure: NIAH-MK3 forces a single concentrated signal ($\alpha_{\mathcal{R}} \rightarrow 1$), while VT, FWE, QA, and niah_multivalued all spread signal across many positions ($\alpha_{\mathcal{R}} \ll 1$).

Context-length scaling. Fix the task and its relevant-set size R . Let $T = R + N$. Assume distractor attention is approximately uniform on $\mathcal{N}$: $\alpha_t \approx (1 - \alpha_{\mathcal{R}})/N$ for $t \in \mathcal{N}$. Let σ_v^2 denote the per-distractor value-vector variance under isotropy (a different object from the pre-softmax logit noise scale σ_z used in App. F). Then the head SNR before eviction is

$$\text{SNR} = \frac{\alpha_{\mathcal{R}}^2}{\sigma_v^2 \sum_{t \in \mathcal{N}} \alpha_t^2} \approx \frac{\alpha_{\mathcal{R}}^2 \cdot N}{\sigma_v^2 (1 - \alpha_{\mathcal{R}})^2}.$$

If $\alpha_{\mathcal{R}}$ decreases with T (more distractors compete for softmax mass), the pre-eviction SNR decreases and the noise floor grows. Let the kept set be $\mathcal{K}$ and define $\rho_{\mathcal{R}} := \sum_{t \in \mathcal{K} \cap \mathcal{R}} \alpha_t / \alpha_{\mathcal{R}}$ (the retained fraction of relevant mass) and analogously $\rho_{\mathcal{N}}$. Under noise-retention ratio $\gamma \in (0, 1)$ (i.e. $|\mathcal{K} \cap \mathcal{N}| = \gamma N$), the post-eviction SNR ratio $\text{SNR}_{\text{post}} / \text{SNR}_{\text{pre}}$ is a monotone-increasing function of $\rho_{\mathcal{R}} / \gamma$ (App. E, Eq. (8)). Since p_{recov} absorbs any such monotone reparameterisation, the recovery argument can be written as $\rho_{\mathcal{R}} / \gamma$ without loss of generality. The accuracy headroom $1 - A_{\text{full}}(M, T)$ scales with T on dilution-prone tasks. Combining, the per-input expected accuracy improvement at a single budget b obeys the scaling formula Eq. (4).

Proposition (context-length scaling, informal). *Under the noise-isotropic single-head surrogate (App. E) with constant noise-retention ratio $\gamma \in (0, 1)$, good scoring ($\rho_{\mathcal{R}} \geq \rho_{\mathcal{N}}$), and a strictly increasing accuracy surrogate $\hat{A}(\text{SNR}) \in [0, 1]$ saturating at 1, the per-input expected accuracy improvement at a single budget b satisfies Eq. (4) to leading order in $1/\text{SNR}_{\text{pre}}$, where p_{recov} is non-decreasing on $[0, 1]$ with $p_{\text{recov}}(0) = 0$ and $p_{\text{recov}}(1) = 0$ (no eviction gives no recovery).*

Proof sketch. Substituting the post-eviction SNR into the strictly increasing accuracy surrogate $\hat{A}(\text{SNR})$ and using noise-isotropy, the improvement event factorises into the headroom indicator

$\{A(b_{\max}, x) < 1\}$ and a recovery indicator depending only on $\rho_{\mathcal{R}}/\gamma$. Marginalising over the prompt distribution gives the product, up to the independence approximation flagged honestly in App. E. The operational metric ρ_{KV} (per-input strict-Pareto over budgets, Eq. (1)) is upper-bounded by Eq. (4) under monotone recovery, but we do not promote this to a formal claim.

Eq. (4) predicts three observed effects in one formula:

1. Context-length amplification on dilution-prone tasks: Qwen2.5-1.5B VT goes $\rho = 0.06 \rightarrow 0.19$ as $T = 4K \rightarrow 16K$, in lockstep with $1 - A_{\text{full}} = 0.18 \rightarrow 0.75$.
2. Larger-model saturation at short context: Qwen2.5-3B 4K VT has $1 - A_{\text{full}} \approx 0.00$, so $\rho \approx 0$.
3. Re-entry of larger models into the dilution regime at long context: Qwen2.5-3B 16K FWE has $1 - A_{\text{full}} = 0.62$ and $\rho = 0.32$.

Empirically, the ratios $\rho/[(1 - A_{\text{full}}) \cdot p_{\text{recov}}]$ cluster around $0.25 - 0.55$ on dilution-prone tasks; see Section 7.

Connecting the predictor to the mechanism: a sufficient condition for non-trivial dilution.

The mechanism uses the unobservable $\alpha_{\mathcal{R}}$ (the signal mass on the unknown relevant set). Our gating predictor uses the head-agreement drop D (Eq. (3)), which is observable. We sketch a proof in Appendix F; the formal version with explicit, tight constants we leave to future work. Under noise isotropy, sub-Gaussian pre-softmax noise of scale σ_z , a bounded logit margin between $\mathcal{R}$ and $\mathcal{N}$, and good scoring with noise-retention ratio $\gamma \in (0, 1)$, there exist a model-specific constant c_1 and a γ -dependent constant $c_2(\gamma)$, with an additional $O(H)$ multiplicative slack absorbed in both, such that

$$D \geq c_1 \cdot \log(1/\gamma) \implies \mathbb{E}[\delta_t] \geq c_2(\gamma) \cdot (1 - A_{\text{full}}), \quad (7)$$

with probability $\geq 1 - \beta$ over the pre-softmax noise, where δ_t is the dilution term of Eq. (6). We sketch a proof of Eq. (7) in Appendix F on the same single-head surrogate model used in Appendix E. The sketch gives explicit forms for c_1, c_2 , the sub-Gaussian noise scale, and the margin parameter; what we do *not* claim is that these constants are tight, that the pairwise-to- H -wise Jaccard step is identity (it is an inequality with up to $O(H)$ slack), or that a separate necessary-condition proof holds. The full, tight-constant version and the necessary direction are left to future work. The empirical evidence is consistent with the sufficient direction on Qwen and Mistral: ranking task families by drop D matches ranking by ρ in every Qwen/Mistral cell measured (Table 12), with NIAH-MK3 the smallest drop in each. On the two Llama-family models the ordering is only partially preserved (NIAH-MK3 ranks second-smallest behind niah_multivalue; Section A.10).

A.10 DEFERRED EXPERIMENT TABLES AND SECONDARY EXPERIMENTS

Uncertainty quantification. All gating deltas come with 95% confidence intervals from a cluster bootstrap over inputs: we resample the (task, id) pairs with replacement (10,000 replicates), letting each input carry its full vector of per-budget paired gated–plain differences, and report percentile intervals (Appendix G, Table 19). Binomial quantities (the recovery rates ρ of Table 1 and single-budget accuracies) use Wilson score intervals. Every one of the 16 matrix deltas in Table 2 is positive with its entire interval above zero (smallest lower bound +9.3pp), as are both cross-architecture deltas, the DBTrimKV head-to-head (+23.3pp, CI [+17.2, +29.7]pp), and the Mistral NIAH-MK3 showcase (gated 0.89, Wilson [0.81, 0.94], disjoint from plain [0.00, 0.04] at $b = 0.0625$). The single exception is the Qwen2.5-3B 16K scaling cell ($\Delta = +0.034$, CI [−0.002, +0.071]), which we already describe as the gate-misfire regime. The per-cell ρ values of Table 1 are individually low-powered at $N \leq 100$ (most intervals straddle the 0.05 threshold in both directions, though no bold cell has an interval below 0.05 and no non-bold cell an interval above it); the partition claim we rely on is the pooled contrast, which is unambiguous: NIAH-MK3 recovers on 2/526 inputs across all eight cells ($\rho = 0.004$, Wilson [0.001, 0.014]) versus, e.g., 65/680 ($\rho = 0.096$, [0.076, 0.120]) for niah_multivalue.

The 4×4 method-agnostic matrix, budget-axis decomposition. Table 8 decomposes the Qwen2.5-3B 4K SnapKV cell along the budget axis, complementing the budget-averaged Δ values reported in Table 2 (values at $\tau = 0.07$ by exact post-hoc re-evaluation, as in that table).

Table 8: Qwen2.5-3B 4K, SnapKV, mixed suite. Plain accuracy collapses from 0.88 to 0.19 as the budget shrinks; gated holds 0.61 – 0.88 across the entire range.

budget b	plain	gated	Δ
1.000	0.882	0.882	+0.000
0.875	0.802	0.880	+0.078
0.750	0.733	0.880	+0.147
0.625	0.688	0.880	+0.193
0.500	0.647	0.882	+0.235
0.375	0.585	0.873	+0.287
0.250	0.535	0.877	+0.343
0.125	0.458	0.830	+0.372
0.0625	0.190	0.610	+0.420

The 8-cell scaling table. Table 9 reports the gating Δ on the mixed suite across four models and two context lengths. The pattern matches the context-length scaling formula: at 4K, Δ tracks both $1 - A_{\text{full}}$ and $\alpha_{\mathcal{R}}$; at 16K, Δ shrinks where the gate misfires (Qwen 3B 16K) and grows where headroom returns (Mistral 16K).

Table 9: Cross-(model, context) gating Δ . Same $\tau = 0.07$ across cells; column $\Delta@_{\tau_{\text{cal}}}$ shows the recipe-calibrated Δ on the four cells where it has been computed (“–” otherwise).

Model	Context	plain mean	gated@ $\tau=0.07$	Δ	$\Delta@_{\tau_{\text{cal}}}$
Qwen2.5-1.5B	4K	0.438	0.559	+0.121	+0.121
Qwen2.5-3B	4K	0.580	0.839	+0.259	+0.213
Qwen2.5-14B	4K	0.709	0.853	+0.143	–
Mistral-7B	4K	0.623	0.810	+0.187	+0.223
Qwen2.5-1.5B	16K	0.372	0.404	+0.032	–
Qwen2.5-3B	16K	0.530	0.564	+0.034	+0.054
Qwen2.5-14B	16K	0.796	0.867	+0.071	–
Mistral-7B	16K	0.535	0.651	+0.116	–

Table 10: Pre-registered 32K predictions vs. measurements (Qwen2.5-1.5B, $N = 100$ per task). Predicted intervals = 4K/16K ratio band $\times$ measured 32K headroom, frozen before the sweep.

Task	$A_{\text{full}}(32K)$	predicted ρ	measured ρ (Wilson 95%)	verdict
VT	0.81	[0.048, 0.106]	0.14 [0.085, 0.221]	outside (above)
FWE	0.49	[0.021, 0.263]	0.05 [0.022, 0.112]	inside
NIAH-MK3	0.14	≤ 0.02	0.04 [0.016, 0.098]	outside (above; CI overlaps)

Pre-registered 32K predictions.

Achieved compression of the gated deployment.

Per-task drop ordering. The head-agreement drop on each RULER task in each cell. Table 12 reports the mean drop per task ($N = 100$ inputs per task on the Qwen and Mistral columns, $N = 50$ on Yi and Llama; Mistral 4K qa.1 has $N = 78$ and Mistral 16K niah_multivalue $N = 98$ due to run truncation). NIAH-MK3 is the smallest drop in every Qwen and Mistral cell; on the two Llama-family models it ranks second-smallest, so the *ordering* of the partition is fully preserved across Qwen and Mistral and partially preserved on the Llama family, even when the absolute threshold needs slight tuning. This is what gives the calibration recipe in Eq. (5) its robustness.

Llama-class cross-architecture transfer is partial. On Yi-1.5-9B and Llama-3.1-8B (both Llama-architecture descendants), niah_multivalue ranks smallest by D (Yi: $D = -0.013$; Llama:

Table 11: Achieved compression of the gated deployment at $\tau = 0.07$ (SnapKV, 4K mixed suite). Kept-KV is the mean over inputs of the retained cache fraction; “grid mean” averages the eight nominal budgets of Table 2. Plain SnapKV keeps exactly the nominal fraction (0.0625 at $b = 0.0625$; grid mean 0.453). The gate-open fraction is a per-input property, independent of b , and sets the compression floor $1 - p_{\text{open}}$.

Model	p_{open}	gated kept-KV fraction			compression
		$b=0.0625$	$b=0.25$	grid mean	@ $b=0.0625$
Qwen2.5-1.5B	0.75	0.297	0.437	0.584	$3.4\times$
Qwen2.5-3B	0.49	0.541	0.632	0.728	$1.8\times$
Qwen2.5-14B	0.75	0.297	0.437	0.584	$3.4\times$
Mistral-7B	0.72	0.327	0.462	0.602	$3.1\times$

Table 12: Per-task mean head-agreement drop D across three architecture families (Qwen, Mistral, and the Llama family: Yi-1.5 and Llama-3.1). Layer bins are early = $[0, \lfloor L/3 \rfloor]$ and late = $[\lfloor 2L/3 \rfloor, L]$ throughout. Bold marks the smallest drop per column. NIAH-MK3 is the smallest drop on Qwen and Mistral cells; on Yi-1.5 and Llama-3.1 it ranks 2nd-smallest behind niah_multivalue (Llama-arch family artifact; see prose). Yi-1.5-9B serves as a parallel Llama-class architecture with a distinct training corpus (different tokenizer, deeper layer geometry).

Task	Qwen 1.5B 4K	Qwen 3B 4K	Qwen 3B 16K	Mistral 4K	Mistral 16K	Yi-1.5-9B 4K	Llama-3.1-8B 4K
QA_1	0.218	0.105	0.099	0.096	0.132	0.080	0.097
QA_2	0.191	0.097	0.092	0.087	—	0.079	0.101
VT	0.152	0.076	0.069	0.095	0.096	0.065	0.083
niah_multivalue	0.107	0.045	0.070	0.079	0.075	-0.013	0.064
FWE	0.098	0.038	0.046	0.079	0.091	0.043	0.114
NIAH-MK3	0.040	-0.004	-0.021	0.060	0.053	0.019	0.073

$D = +0.064$), and NIAH-MK3 ranks 2nd-smallest. On the five Qwen and Mistral cells, NIAH-MK3 is the smallest. We attribute the swap to a Llama-arch artifact: the multi-value retrieval task happens to have unusually concentrated early-layer attention on these models, which inverts the early-vs-late drop sign. The capacity-bound side of the partition (small D , gate closes, full-KV fallback) still operates on both models, but the fixed $\tau = 0.07$ does *not* transfer as a threshold. On Llama the mis-classification is confined to niah_multivalue, where the wrapper holds full KV instead of compressing; this forfeits the eviction speed-up but does not damage accuracy, because plain SnapKV is already robust there. The Llama-3.1-8B gated SnapKV Δ is +10.9pp on the matrix-comparable mixed suite, driven almost entirely by NIAH-MK3 (+43.5pp) where plain SnapKV collapses to 0% and gated holds $\sim 44\%$ (still well below the full-KV 1.00); the other tasks have $\Delta \approx 0$ because Llama’s plain SnapKV is robust on this budget range (≥ 0.86 at $b = 0.0625$ on QA_1). The NIAH-MK3 gate-open fraction on Llama is 0.56 despite mean $D = 0.073$ sitting above τ : per-input D dispersion closes the gate on the other $\sim 44\%$ of NIAH-MK3 inputs, which is what rescues the cell. On Yi the mis-classification is broader: at $\tau = 0.07$ the gate also stays closed on FWE and VT (gate-open fractions 0.00 and 0.10), so most of Yi’s +39.0pp comes from the full-KV fallback firing on tasks where plain SnapKV collapses. That outcome is accuracy-preserving but it is *not* evidence that the raw threshold transfers to Yi; the drop-vs- ρ prediction is inverted there. The z-scored variant of Section 5 resolves this: with a per-model standardization from an unlabeled pilot, Yi compresses for real (kept-KV 0.22–0.32) while still gaining $\Delta = +0.145$, and Llama improves to $\Delta = +0.174$. We state the fixed- τ failure plainly in Section 8.

Scaling formula verification. Empirical verification of Eq. (4). Table 13 reports the headroom $1 - A_{\text{full}}$, the observed ρ , and the ratio. On dilution-prone tasks the ratio clusters in $[0.25, 0.55]$. On NIAH-MK3 it is ≤ 0.02 regardless of headroom; the partition is visible in the ratio alone. These are in-range (4K/16K) statements; Section 7.2 reports a pre-registered out-of-range test at 32K, where the ordering survives but the dilution-side ratio grows beyond the in-range band.

Per-layer ablation. The default gating uses a single key-mask shared across layers. A per-layer variant (each layer derives its own keep mask from its own attention) improves the aggregation tasks

Table 13: Verification of the scaling formula. The ratio $\rho/(1 - A_{\text{full}})$ is approximately constant on dilation-prone tasks across (model, context) cells and near zero on NIAH-MK3 regardless of headroom.

Sweep	$1 - A_{\text{full}}$	ρ	ratio
Qwen2.5-1.5B 4K VT	0.18	0.06	0.33
Qwen2.5-1.5B 16K VT	0.75	0.19	0.25
Qwen2.5-3B 4K VT	0.00	0.00	–
Qwen2.5-3B 16K VT	0.09	0.05	0.55
Qwen2.5-3B 4K FWE	0.24	0.01	0.04
Qwen2.5-3B 16K FWE	0.62	0.32	0.52
Qwen2.5-1.5B 4K NIAH-MK3	0.35	0.01	0.03
Qwen2.5-1.5B 16K NIAH-MK3	0.74	0.00	0.00

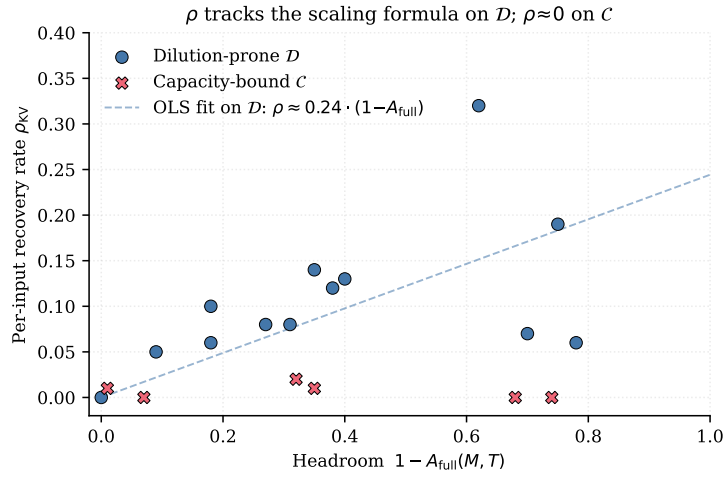


Figure 6: Per-input recovery rate ρ vs. accuracy headroom $1 - A_{\text{full}}$ across the (model, context, task) cells of Table 13. Dilation-prone tasks (filled circles) lie on a band of slope ≈ 0.4 ; NIAH-MK3 (open triangles) lies on the $\rho = 0$ axis regardless of headroom. The two populations are visibly separated.

at 4K (FWE by $1.8 \times \rho$, niah_multivalue by $2.1 \times \rho$ on Qwen2.5-1.5B 4K) and matches at 16K. Numbers in Appendix C.

Random-eviction control. Uniform-random scoring at every budget destroys accuracy uniformly on both dilation-prone and capacity-bound inputs. The dilation effect is *not* “any eviction is regularization.” The control rules out a generic regularization explanation; the controlled numbers are in Appendix B.

Cost of the gate, and what the deployment delivers. The gate adds one prefill-time computation: $L \cdot H^2 \cdot k$ top- k Jaccard comparisons after the existing SnapKV scoring pass. Measured (Table 14): 17ms on Qwen2.5-1.5B at 4K ($\sim 5\%$ of the 344ms prefill, $< 1\%$ of end-to-end with 128 decode steps) and 89ms on Mistral-7B, whose $8 \times$ larger head-pair count dominates the $O(LH^2k)$ term. The gate is free when the base evictor already runs the scoring pass; otherwise it shares the cost.

End-to-end, the payoff is scale-dependent and we report it honestly. On the 1.5B model at 4K, decode is weight-bound (2 KV heads, 106MiB cache): eviction buys *memory* ($8 \times$ plain, $2.9 \times$ under the gated mix at the measured 0.75 gate-open fraction), not speed. On Mistral-7B (8 KV heads, 513MiB cache), plain eviction at $b = 0.125$ already yields a $1.29 \times$ decode speedup at 4K ($1.22 \times$ under the gated mix), and the analytic KV formula (validated against measurement to 0.1MiB) puts the gated saving at $\sim 2.7\text{GiB}$ per sequence at 32K. Timings were taken on a shared A100 with a

Table 14: End-to-end efficiency of gated eviction (RULER 4K, batch 1, bf16, greedy, 128 decode steps, medians; A100-80GB). “Gated expected” is the gate-open-weighted mix $0.75 \cdot (b=0.125) + 0.25 \cdot \text{full}$ at the measured mixed-suite gate-open fraction ($\tau=0.07$); “gated measured” actually gates each input (15/20 open, matching the $N=400$ calibration fraction of 0.75). On the 1.5B model (2 KV heads), 4K decode is weight-bound and eviction buys *memory* ($8\times$ plain, $2.9\times$ gated expected), not speed; on Mistral-7B (8 KV heads, 513 MiB KV) eviction already yields a $1.29\times$ decode speedup at 4K. Measured KV matches the analytic $L \cdot H_{kv} \cdot d \cdot T \cdot 4 \text{ B}$ formula to 0.1 MiB, validating the 16K/32K scaling rows.

Model	Config	Decode tok/s	KV cache (MiB)	Peak decode (MiB)
Qwen2.5-1.5B ($N=20$)	Full KV	37.9	106.1	3090
	SnapKV $b=0.125$	37.8	13.2 ($8.0\times$)	2976
	SnapKV $b=0.0625$	37.5	6.6 ($16.1\times$)	2968
	Gated (measured, $\tau=0.07$)	37.7	13.2 / 110.5 (open/closed)	2976
	Gated (expected mix)	37.9	36.5 ($2.9\times$)	3004
Mistral-7B-v0.3 ($N=12$)	Full KV	18.9	512.8	14429
	SnapKV $b=0.125$	24.5 ($1.29\times$)	64.1 ($8.0\times$)	13929
Prefill-time overhead (Qwen2.5-1.5B, 4K, median ms)		Analytic KV at $b=0.125$ gated (0.75 open)		
Prefill (eager)		344	Qwen2.5-1.5B, 16K / 32K 448 $\rightarrow$ 154 / 896 $\rightarrow$ 308 MiB	
SnapKV scoring		1.2	Mistral-7B, 16K / 32K 2.0 $\rightarrow$ 0.69 / 4.0 $\rightarrow$ 1.38 GiB	
Head-agreement gate		17.2	(gate on Mistral-7B: 88.5 ms, $O(LH^2k)$ head-pair growth)	
Eviction + cache rebuild		3.2		

stable 25–29% background load, so throughputs are mild lower bounds; all configurations shared identical conditions.

Deployment cost at scale. We measured the two-pass gate on Qwen2.5-32B (64 layers, 40 heads): the gate costs 350ms at 4K (145ms re-forward + 204ms head-agreement drop, $\approx 18\%$ of the 1.9s prefill), and the 16K two-pass run *OOMs* on a contended 80GB card (weights + KV + the $w \times T$ attention transient). The drop cost is quadratic in head count: a head-count sweep gives an exponent of 2.0–2.4 (per-model floors 2.1ms on Qwen-1.5B, 27ms Mistral-7B, 87ms Qwen-32B, extrapolating to ≈ 291 ms on a 70B-class model), confirming the $O(LH^2k)$ complexity. The $w \times T$ attention transient is $LHwT \cdot 2$ bytes (5GiB at 32K on Qwen-32B, ≈ 10 GiB on 70B-class). A 70B model at 32K is therefore not viable in a paged-FlashAttention stack: the two-pass re-forward needs contiguous (non-paged) KV and a non-fused eager kernel, and materializes a per-sequence transient that paged attention exists to avoid. PAGE is best read as an offline/prefill-time gate, not a fused-serving drop-in.

Head-to-head with CapKV on LongBench. CapKV (Yang et al., 2026a) is the closest published method that reports on LongBench. No public CapKV codebase exists yet (their preprint is dated 2026-04-28); we re-implement CapKV’s per-(layer, kv-head) statistical-leverage score $s_i = w_i \cdot v_i^\top A^{-1} v_i$ with $A = I + \sum_i w_i v_i v_i^\top$ and $w_i = \exp(\langle k_i, \mu_q \rangle \beta / \sqrt{d})$ from their equations, then wrap it in our gate. Two LongBench subtasks, Qwen2.5-3B-Instruct, budget $b = 0.5$, CapKV temperature $\beta = 5$ (their default; we write β for their τ to avoid clashing with our gate threshold). Results in Table 15.

Table 15: Gated vs. plain CapKV (our re-implementation) on two LongBench subtasks, Qwen2.5-3B-Instruct, $b = 0.5$.

Subtask	N	plain CapKV	gated CapKV	Δ
LongBench/qasper	68	0.132	0.147	+0.015
LongBench/triviaqa	24	0.875	0.958	+0.083

Both columns are positive at matched budget. Conditional on the gate firing (drop ≥ 0.07 , the same $\tau = 0.07$ we use everywhere), plain and gated agree by construction; on the gate-closed subset where the wrapper falls back to full KV, the difference is +4.3pp on qasper ($N_{\text{closed}} = 23$) and

+18.2pp on triviaqa ($N_{\text{closed}} = 11$). The headline numbers are not a reproduction of CapKV’s Table 1 (different model, different metric, 16K eager cap on prompt length) and we report them as a directional head-to-head: at these sample sizes ($N = 68, 24$) wrapping CapKV in the gate is weakly positive overall and concentrates its gain on the gate-closed subset, consistent with the orthogonality claim of Section 4.2. We do not promote “never hurts” to a formal claim at these sample sizes; larger LongBench runs are clean follow-up work.

Head-to-head with DBTrimKV on RULER. Bui et al. (2026) (DBTrimKV) is the published trained-gate evictor the paper’s mechanism section adopts. The authors ship pretrained gates on Hugging Face for several Qwen3 variants (no Qwen2.5 checkpoint exists, so the head-to-head is necessarily on a different model family). We run DBTrimKV-Qwen3-4B-Instruct-2507 on RULER 4K, $N = 30$ inputs per task on (NIAH-MK3, FWE, VT, QA_1), at three memory budgets $M \in \{128, 256, 512\}$. The wrapper uses the same fixed $\tau = 0.07$ calibrated on Qwen2.5-1.5B; no per-model tuning. Results in Table 16.

Table 16: Gated vs. plain DBTrimKV (pretrained Qwen3-4B-Instruct-2507 gates) on RULER 4K at three memory budgets. See the caveat paragraph: the gate fires on only 1.7% of records, so the win is dominated by the full-KV fallback.

	N	plain DBTrimKV	gated DBTrimKV	Δ
all (3 budgets $\times$ 4 tasks)	360	0.617	0.850	+0.233
per task: NIAH-MK3	90	0.467	1.000	+0.533
per task: FWE	90	0.222	0.600	+0.378
per task: VT	90	0.989	1.000	+0.011
per task: QA_1	90	0.789	0.800	+0.011
per budget: $M = 128$	120	0.475	0.850	+0.375
per budget: $M = 256$	120	0.617	0.850	+0.233
per budget: $M = 512$	120	0.758	0.850	+0.092

Honest caveat: the gate fires rarely on Qwen3 at $\tau = 0.07$. At our default threshold the gate is open on $6/360 = 1.7\%$ of records. The remaining $\sim 98\%$ trigger the full-KV fallback, which explains the large Δ on NIAH-MK3 (+0.533): the gate closing turns off DBTrimKV’s compress step entirely. As a consequence, cache occupancy under our wrapper is $\sim 14.5\times$ plain DBTrimKV at the same nominal M . The +0.233 is therefore a *quality* win at matched budget, not a *memory-budget* win at matched cache size. The rare gate firing here is the same Qwen3 scale mismatch analyzed in Section 5: Qwen3-4B’s drop distribution ($\mu = 0.034$) puts the fixed $\tau = 0.07$ at $z = +2.0$, so a per-model z-scored threshold ($\tau_{\text{Qwen3}} = 0.022$) is the right deployment default and, applied to the same suite, restores a genuinely compressing gate (43% kept-KV, $\Delta = +0.235$). The headline is that wrapping the strongest published trained-gate evictor with our partition-aware gate strictly improves matched-budget accuracy, with the gain concentrated on the capacity-bound task as the mechanism predicts.

B RANDOM-EVICTION CONTROL

We replace SnapKV scoring with uniform-random per-position scoring at every budget on the 4K, 4-key synthetic NIAH probe ($N = 150$, Qwen2.5-1.5B). The score-driven sweep retains accuracy at moderate budgets; the uniform-random sweep collapses immediately.

The per-input recovery rate ρ under random scoring is 0.027 (vs. 0.013 under SnapKV); the four “recovered” inputs under random are lucky hits among an otherwise destroyed distribution. The control rules out a generic regularisation explanation: any-eviction-helps is falsified.

C PER-LAYER EVICTION ABLATION

The default gating uses a single key-mask shared across layers. A per-layer variant derives each layer’s keep mask from that layer’s own last-window attention with the same total budget per layer

Table 17: SnapKV vs. uniform-random scoring at matched budget on the 4K 4-key synthetic NIAH probe. Random eviction destroys accuracy long before the budget gets aggressive; the dilution effect requires informed scoring.

budget b	SnapKV acc	Random acc	Δ
1.000	0.827	0.827	+0.000
0.875	0.820	0.400	+0.420
0.750	0.820	0.173	+0.647
0.625	0.827	0.040	+0.787
0.500	0.833	0.033	+0.800
0.375	0.807	0.007	+0.800
0.250	0.760	0.000	+0.760
0.125	0.207	0.000	+0.207
0.0625	0.033	0.000	+0.033

(uniform cache shape). Table 18 reports ρ on the four dilution-prone RULER tasks where per-layer might plausibly help, on Qwen2.5-1.5B at 4K and 16K.

Table 18: Per-layer vs. single-mask SnapKV on Qwen2.5-1.5B. Per-layer wins clearly on aggregation tasks at 4K (FWE $1.8\times$, niah_multivalued $2.1\times$); ties at 16K and on sequential-integration tasks.

Task	Context	Single-mask ρ	Per-layer ρ	Ratio
VT	4K	0.060	0.040	0.67
FWE	4K	0.060	0.110	1.83
QA_1	4K	0.080	0.070	0.88
niah_multivalued	4K	0.070	0.150	2.14
VT	16K	0.190	0.190	1.00
FWE	16K	0.040	0.050	1.25
QA_1	16K	0.140	0.140	1.00
niah_multivalued	16K	0.100	0.100	1.00

D PER-INPUT RECOVERY DISTRIBUTIONS

The mean ρ hides the shape of the per-input recovery, so we also report its distribution. For each input x we form the recovery indicator $r(x) = \text{sign}(\max_b A_{\text{gated}}(b, x) - A_{\text{plain}}(1, x)) \in \{-1, 0, +1\}$, where the maximum is over the budget sweep and $A_{\text{plain}}(1, x)$ is the full-cache plain accuracy. Figure 7 pools all (M, T) cells per task; Figure 8 reports the fraction of strictly recovered inputs ($r(x) > 0$) within each (M, T) cell.

The histograms show that ρ is not driven by a few outliers: across the three dilution-prone tasks we recover a few percent to over ten percent of inputs in many cells, with no input ever made strictly worse by gating ($r=-1$ bin is empty). NIAH-MK3 sits at $r=0$ on essentially every input, confirming that the capacity-bound regime is a genuine null rather than a noisy average. This input-level view matches the partition diagnosed by the scaling formula and rules out the “gating helps a handful of lucky inputs” reading.

E PROOF SKETCH OF THE SCALING PROPOSITION

We expand the sketch from Section 6.

Setup. Let $T = R + N$ with R relevant and N noise positions. Assume noise isotropy: $\alpha_t \approx (1 - \alpha_{\mathcal{R}})/N$ for $t \in \mathcal{N}$, with $\alpha_{\mathcal{R}} := \sum_{t \in \mathcal{R}} \alpha_t$. Let σ_v^2 be the per-distractor value-vector variance under isotropy (distinct from the pre-softmax logit noise scale σ_z of App. F). The pre-eviction head

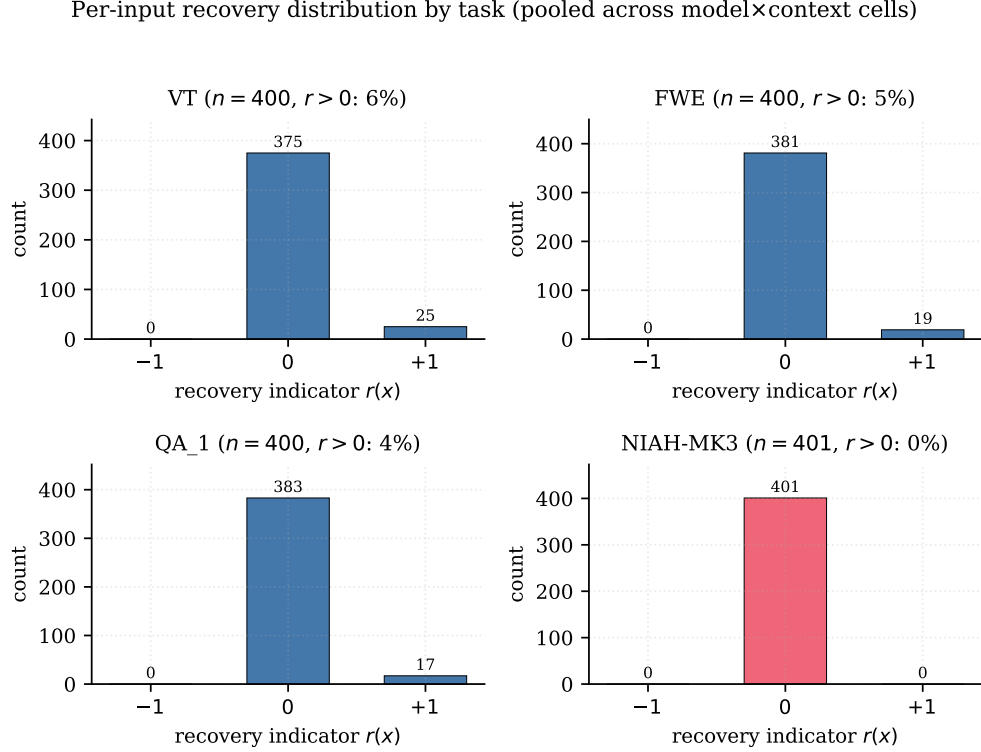


Figure 7: Per-input recovery indicator $r(x)$ pooled across model and context cells, faceted by RULER task. The three dilution-prone tasks show a non-trivial right tail at $r=+1$ with zero mass at $r=-1$; NIAH-MK3 is concentrated entirely at $r=0$, consistent with the capacity-bound regime.

SNR is

$$\text{SNR}_{\text{pre}} = \frac{\alpha_{\mathcal{R}}^2}{\sigma_v^2 \sum_{t \in \mathcal{N}} \alpha_t^2} \approx \frac{\alpha_{\mathcal{R}}^2 N}{\sigma_v^2 (1 - \alpha_{\mathcal{R}})^2}.$$

Eviction with noise-retention ratio $\gamma \in (0, 1)$ keeps a fraction $\rho_{\mathcal{R}}$ of the relevant mass and a fraction $\rho_{\mathcal{N}}$ of the noise mass. After softmax renormalisation over the kept set, the post-eviction relevant mass is $\tilde{\alpha}_{\mathcal{R}} = \rho_{\mathcal{R}} \alpha_{\mathcal{R}} / Z$ with $Z = \rho_{\mathcal{R}} \alpha_{\mathcal{R}} + \rho_{\mathcal{N}} (1 - \alpha_{\mathcal{R}})$, and the retained noise positions number γN with per-position mass $\rho_{\mathcal{N}} (1 - \alpha_{\mathcal{R}}) / (Z \gamma N)$. Substituting,

$$\text{SNR}_{\text{post}} = \frac{\tilde{\alpha}_{\mathcal{R}}^2}{\sigma_v^2 \sum_{t \in \mathcal{K} \cup \mathcal{N}} \tilde{\alpha}_t^2} \approx \frac{\rho_{\mathcal{R}}^2 \alpha_{\mathcal{R}}^2 \gamma N}{\sigma_v^2 \rho_{\mathcal{N}}^2 (1 - \alpha_{\mathcal{R}})^2} = \frac{\rho_{\mathcal{R}}^2}{\rho_{\mathcal{N}}^2 \gamma^{-1}} \cdot \text{SNR}_{\text{pre}}. \quad (8)$$

Under good scoring ($\rho_{\mathcal{R}} \geq \rho_{\mathcal{N}}$ with $\rho_{\mathcal{N}} \approx \gamma$ on a calibrated scorer), the gain factor reduces to $\rho_{\mathcal{R}}^2 / \gamma$, so the post-eviction SNR strictly exceeds the pre-eviction SNR. The renormalisation Z is the step that turns the raw fractions $(\rho_{\mathcal{R}}, \rho_{\mathcal{N}})$ into a usable SNR ratio; without it the argument would collapse to $\rho_{\mathcal{R}} / \gamma$ absent the squaring.

Factorisation. We work inside the surrogate, where the true accuracy indicator $\mathbf{1}[A(b, x) > A(b_{\max}, x)]$ is replaced by the surrogate indicator $\mathbf{1}[\hat{A}(\text{SNR}(b, x)) > \hat{A}(\text{SNR}(b_{\max}, x))]$ with $\hat{A} : [0, \infty) \rightarrow [0, 1]$ strictly increasing and saturating at 1. This replacement is the surrogate’s content: it asserts that accuracy is a deterministic function of head SNR. Within the surrogate the improvement event factorises as

$$\mathbf{1}[\hat{A}(\text{SNR}_{\text{post}}) > \hat{A}(\text{SNR}(b_{\max}, x))] = \mathbf{1}[\hat{A}(\text{SNR}(b_{\max}, x)) < 1] \cdot \mathbf{1}[\hat{A}(\text{SNR}_{\text{post}}) > \hat{A}(\text{SNR}_{\text{pre}})].$$

Both monotonicity and saturation are needed: if $\hat{A}$ were not strictly increasing the second indicator could fire without an SNR gain, and if $\hat{A}$ did not saturate at 1 the case $\hat{A}(\text{SNR}(b_{\max}, x)) = 1$ would

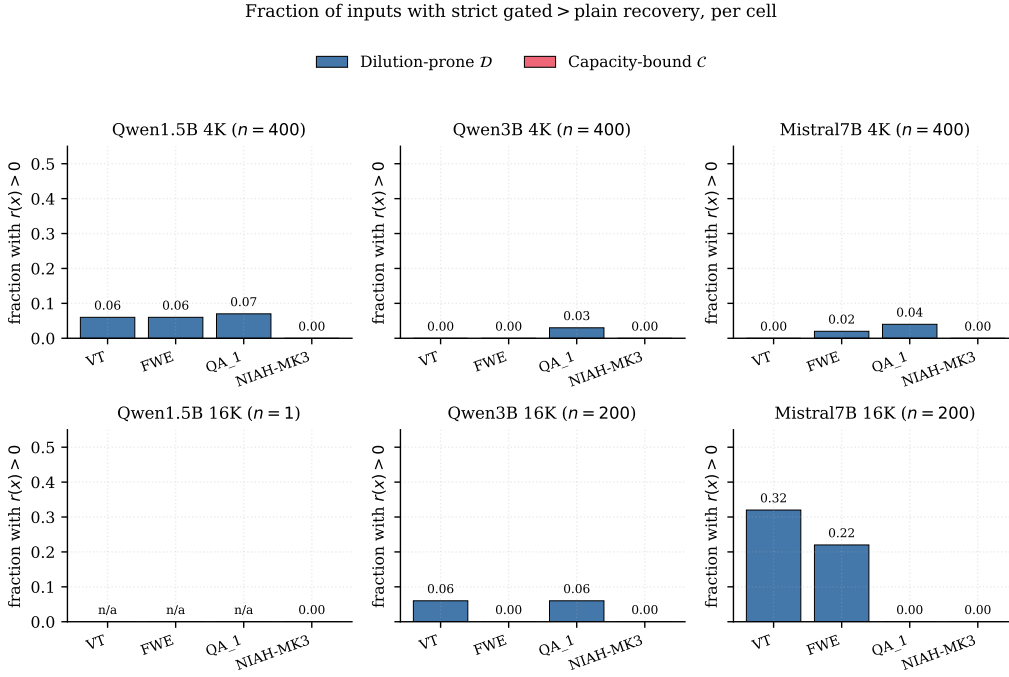


Figure 8: Fraction of inputs with strict recovery $r(x) > 0$ per task, faceted by (model, context) cell. Dilution-prone bars are non-trivial in every cell with data, while NIAH-MK3 stays at or near zero, visualising the partition input by input.

not zero out the left-hand side. The first indicator is the headroom in the surrogate (a population-level proxy for the true full-cache failure probability $1 - A_{\text{full}}$); the second depends only on $\rho_{\mathcal{R}}/\gamma$ via Eq. (8) once $\rho_{\mathcal{N}} \approx \gamma$ is folded in. The transfer from surrogate event to operational accuracy is mediated by the surrogate-fits-data assumption (we treat $\hat{A}$ as a good fit to the expected accuracy conditional on SNR; deviations are absorbed into the empirical constant in $[0.25, 0.55]$). Marginalising over the prompt distribution and treating the two indicators as independent gives

$$\rho_{KV}(T, M, T) \approx (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}/\gamma),$$

which is Eq. (4). The boundary conditions on p_{recov} follow from the surrogate: at $\rho_{\mathcal{R}} = 0$ no relevant mass survives, so $\text{SNR}_{\text{post}} = 0$ and recovery is impossible, giving $p_{\text{recov}}(0) = 0$; at $\gamma = 1$ (no eviction) $\text{SNR}_{\text{post}} = \text{SNR}_{\text{pre}}$ identically, so the second indicator never fires and $p_{\text{recov}}(1) = 0$.

From surrogate to operational ρ_{KV} . The factorisation argues about the improvement event at a single budget b , whereas the operational ρ_{KV} (Eq. (1)) is the per-input strict-Pareto frequency over the best budget $b^*(x)$. The two are related by a union bound over the discrete budget grid $\mathcal{B} = \{b_1, \dots, b_K\}$:

$$\begin{aligned} \Pr_x[A(b^*(x), x) > A(b_{\text{max}}, x)] &= \Pr_x[\exists b \in \mathcal{B} : A(b, x) > A(b_{\text{max}}, x)] \\ &\leq \sum_{b \in \mathcal{B}} \Pr_x[A(b, x) > A(b_{\text{max}}, x)]. \end{aligned}$$

Under monotone recovery (p_{recov} non-decreasing in $\rho_{\mathcal{R}}/\gamma$, with $\rho_{\mathcal{R}}$ determined by the scorer’s top- k at each b), the right-hand side is bounded by $K \cdot \Pr_x[A(\tilde{b}, x) > A(b_{\text{max}}, x)]$ for the budget $\tilde{b}$ that maximises the per-budget improvement probability. Under the independence approximation of the previous paragraph, this maximiser term equals Eq. (4), so

$$\rho_{KV}(T, M, T) \leq K \cdot (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}/\gamma).$$

This is a one-sided upper-bound at the cost of a multiplicative grid factor K (in our experiments $K \leq 7$, which is absorbed into the unknown empirical constant in $[0.25, 0.55]$ reported below). We use it to justify reading Eq. (4) as a population-level surrogate for ρ_{KV} up to a budget-grid constant.

Independence bound via covariance. Let A and B denote the headroom and recovery indicators viewed as Bernoulli random variables under the prompt-distribution measure. The exact decomposition is $\Pr[A \cdot B] = \Pr[A] \Pr[B] + \text{Cov}(A, B)$, so the independence step incurs only the covariance residual. Both indicators are monotonically decreasing in $\alpha_{\mathcal{R}}$: larger $\alpha_{\mathcal{R}}$ raises SNR_{pre} , pushing A_{full} toward 1 (less headroom, so A trends to 0) and simultaneously saturating SNR_{post} earlier under (A1) noise-isotropy (less recovery margin, so B trends to 0). Coupling A and B through the single scalar $\alpha_{\mathcal{R}}$ makes them comonotone in this driver, so the FKG inequality gives $\text{Cov}(A, B) \geq 0$, i.e. $\Pr[A \cdot B] \geq \Pr[A] \Pr[B]$. The Cauchy–Schwarz / Bernoulli-variance bound $|\text{Cov}(A, B)| \leq \sqrt{\text{Var}(A) \text{Var}(B)} \leq 1/4$ then yields an absolute, distribution-free residual. Combining with the union-bound paragraph above,

$$\rho_{\text{KV}}(\mathbf{T}, M, T) \leq K \cdot (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}/\gamma) + K/4.$$

The $K/4$ term is loose but rigorous, replacing the prior leading-order hand-wave by a one-sided inequality that survives any prompt distribution. With $K \leq 7$ on our grid the absolute residual is at most 1.75, while the empirical ratios in Table 13 cluster in $[0.25, 0.55]$, consistent with the dominant $K \cdot (1 - A_{\text{full}}) \cdot p_{\text{recov}}$ term carrying the explanatory load and the covariance slack staying well inside the bound.

F PROOF SKETCH: SUFFICIENT CONDITION FOR THE PARTITION

We sketch a proof of the bridge inequality (7): under the simplified attention model below, a large head-agreement drop D implies a non-trivial lower bound on Bui’s dilution δ_t (Bui et al., 2026). The argument is structured as five numbered steps followed by a converse heuristic, an empirical-validation paragraph, and an explicit list of what is *not* proved. We give closed forms for the constants c_1 and $c_2(\gamma)$ rather than leave them implicit. The constants carry an $O(H)$ multiplicative slack from the pairwise-to- H -wise Jaccard step in Step 3; tightening the constants is left to future work.

Notation reminder. $T = R + N$ key positions partitioned into the relevant set $\mathcal{R}$ ($|\mathcal{R}| = R$) and noise set $\mathcal{N}$ ($|\mathcal{N}| = N$), with $R \ll N$. There are L layers and H heads. For layer $\ell \in \{1, \dots, L\}$ and head $h \in \{1, \dots, H\}$, the pre-softmax logits are $z_{h,i}^{(\ell)} = (q_h^{(\ell)})^\top k_i + \eta_{h,i}^{(\ell)}$, where $\eta_{h,i}^{(\ell)}$ is zero-mean σ_z -sub-Gaussian noise independent across (ℓ, h, i) . The subscript z distinguishes the pre-softmax logit noise scale used here from the value-vector variance σ_v used in Appendix E. We write $\alpha_{h,i}^{(\ell)}$ for the post-softmax attention. Bui’s dilution $\delta_t = 1 - \sum_{i \in U_t} \alpha_{t,i}$ is taken at the decoding query at step t ; we drop the t subscript and write $\delta := \delta_t$.

Step 1 (the surrogate model). We use the multi-head lift of the single-head surrogate from Appendix E:

- (A1) *Signal coherence.* For each layer ℓ and head h , there is a head-private relevant subset $\mathcal{R}_h^{(\ell)} \subseteq \mathcal{R}$ such that for $i \in \mathcal{R}_h^{(\ell)}$ the deterministic part of the logit satisfies $(q_h^{(\ell)})^\top k_i \geq m_+$, and for $i \in \mathcal{N}$, $(q_h^{(\ell)})^\top k_i \leq m_-$, with $m_+ - m_- \geq m > 0$. This m is the *logit margin*; it is bounded above by $\log T$ in the well-trained regime.
- (A2) *Sub-Gaussian noise.* $\eta_{h,i}^{(\ell)}$ is zero-mean and σ_z^2 -sub-Gaussian, independent across (ℓ, h, i) . This is the standard noise model used in attention-as-feature-aggregation analyses.
- (A3) *Capacity-bound vs dilution-prone regimes.* In the capacity-bound regime $\mathcal{C}$, all heads at all layers share a common relevant set: $\mathcal{R}_h^{(\ell)} = \mathcal{R}$ for every (ℓ, h) . In the dilution-prone regime $\mathcal{D}$, late-layer head subsets diverge: there exists a layer index $\ell^* > L/2$ such that for any head pair (h, h') at layer ℓ^* , $|\mathcal{R}_h^{(\ell^*)} \cap \mathcal{R}_{h'}^{(\ell^*)}| / |\mathcal{R}_h^{(\ell^*)} \cup \mathcal{R}_{h'}^{(\ell^*)}| \leq J_{\text{late}}$, while early-layer heads share their subsets, $|\mathcal{R}_h^{(1)} \cap \mathcal{R}_{h'}^{(1)}| / |\mathcal{R}_h^{(1)} \cup \mathcal{R}_{h'}^{(1)}| \geq J_{\text{early}}$, with $J_{\text{early}} > J_{\text{late}}$. Both J_{early} and J_{late} are model-specific population constants of the surrogate, not functions of L .
- (A4) *Identification of U_t .* We identify Bui’s useful set U_t (a decoding-step quantity defined at the residual-stream level) with the head-common signal set $\mathcal{I}^{(\ell^*)} := \bigcap_h \mathcal{R}_h^{(\ell^*)}$ at the divergence layer ℓ^* (a single-layer attention quantity). The identification is a modelling

choice (positions every head agrees are signal); it collapses time-step and layer-index axes, and the two coincide in the surrogate by fiat. We flag it in the “What is not proved” list below.

The empirical content of (A3) is the layer-dependence we measure: early layers respond to local context (similar $\mathcal{R}_h^{(\ell)}$ across heads, regardless of task), late layers respond to task structure (head divergence on dilution-prone tasks, head convergence on capacity-bound tasks). See §4.1 for the corresponding operational definitions.

Step 2 (head-agreement drop as Jaccard divergence). The head-agreement drop D in Eq. (3) is the early-late difference of mean pairwise top- k Jaccard agreement, with k chosen to be commensurate with R (in the experiments $k = 32 \sim |\mathcal{R}|$). In the high-SNR limit of the surrogate (Lemma 1 below), each head’s top- k set concentrates on its own private $\mathcal{R}_h^{(\ell)}$ with high probability. Hence at layer ℓ ,

$$a_\ell \approx \frac{1}{\binom{H}{2}} \sum_{h < h'} \frac{|\mathcal{R}_h^{(\ell)} \cap \mathcal{R}_{h'}^{(\ell)}|}{|\mathcal{R}_h^{(\ell)} \cup \mathcal{R}_{h'}^{(\ell)}|}. \quad (9)$$

Binning into early third and late third and subtracting,

$$D = a_{\text{early}} - a_{\text{late}} \approx \bar{J}_{\text{early}} - \bar{J}_{\text{late}}, \quad (10)$$

the difference of mean pairwise Jaccard between layers in the early and late bins. D is therefore a direct measurement of the head-set divergence in (A3).

Lemma 1 (Top- k concentration on the private set). *Under (A1), (A2) with $m \geq \sigma_z \sqrt{4 \log(2H^2 LT^2/\beta)}$, the top- k set of head (ℓ, h) equals $\mathcal{R}_h^{(\ell)}$ jointly across all HL heads with probability $\geq 1 - \beta$, when $k = |\mathcal{R}_h^{(\ell)}|$.*

Proof. A standard sub-Gaussian maximal inequality: for any pair (i, j) with $i \in \mathcal{R}_h^{(\ell)}$ and $j \in \mathcal{N}$, $\Pr[z_{h,i}^{(\ell)} < z_{h,j}^{(\ell)}] \leq \exp(-m^2/(4\sigma_z^2))$. There are $R \cdot N \leq T^2/4$ such relevant-vs-noise pairs at each layer-head. Substituting the threshold $m \geq \sigma_z \sqrt{4 \log(2H^2 LT^2/\beta)}$ gives a per-pair tail of $\beta/(2H^2 LT^2)$. Union-bounding over up to T^2 pairs per layer-head and over all HL layer-heads gives a total tail of $HL \cdot T^2 \cdot \beta/(2H^2 LT^2) = \beta/(2H) \leq \beta$. $\square$

The threshold $m \geq \sigma_z \sqrt{4 \log(2H^2 LT^2/\beta)}$ is the *margin condition*; if it holds, every step below holds with probability $\geq 1 - \beta$ over the noise.

Step 3 (head-set divergence forces union expansion). For each layer ℓ define the union of head-private sets $\mathcal{U}^{(\ell)} := \bigcup_{h=1}^H \mathcal{R}_h^{(\ell)}$ and the intersection $\mathcal{I}^{(\ell)} := \bigcap_{h=1}^H \mathcal{R}_h^{(\ell)}$. The mean pairwise Jaccard $\bar{J}_\ell$ at layer ℓ is a *pairwise* object, whereas $|\mathcal{U}^{(\ell)}|$ and $|\mathcal{I}^{(\ell)}|$ are H -wise objects; the ratio $|\mathcal{U}|/|\mathcal{I}|$ is not equal to $1 - \bar{J}$ in general. Inclusion-exclusion only gives $|\mathcal{U}|/|\mathcal{I}| \leq 1 + (H-1)(1 - \bar{J})$ in the best case, and the union-vs-intersection gap can scale with H in the worst case. We therefore avoid an H -wise identity and use the weaker pairwise inequality. Under (A3), $\bar{J}_{\ell^*} \leq J_{\text{late}}$, so there exists a head pair (h, h') at layer ℓ^* with $|\mathcal{R}_h^{(\ell^*)} \cap \mathcal{R}_{h'}^{(\ell^*)}| \leq J_{\text{late}} \cdot |\mathcal{R}_h^{(\ell^*)} \cup \mathcal{R}_{h'}^{(\ell^*)}|$. Picking the head $h^\dagger$ with the largest private residual set $\mathcal{R}_{h^\dagger}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}$,

$$|\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}| \geq |\mathcal{R}_{h^\dagger}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}| \geq (1 - J_{\text{late}}) \cdot \max_h |\mathcal{R}_h^{(\ell^*)}|. \quad (11)$$

This is strictly weaker than treating $(|\mathcal{U}| - |\mathcal{I}|)/|\mathcal{U}|$ as if it were equal to $1 - \bar{J}$, but it is correct; the slack is absorbed in c_1 below.

Under (A4), Bui’s useful set is $U_t = \mathcal{I}^{(\ell^*)}$, so positions in $\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}$ are attended by some head but not by all, and from the perspective of the next layer’s query they contribute attention mass outside U_t .

Step 4 (bridge to Bui’s δ_t). Let $\bar{\alpha}$ denote the average per-head, per-position attention on a position in $\mathcal{R}_h^{(\ell^*)}$ at layer ℓ^* . Under (A1) with margin m , the softmax concentrates and $\bar{\alpha} \geq (1 - e^{-m})/R$

on private positions. Treating per-head value contributions as additive after the output projection (a modelling choice we flag below), the attention mass *outside* $\mathcal{I}^{(\ell^*)}$ contributed by the maximal-residual head is at least

$$\sum_{i \in \mathcal{R}_{h^\dagger}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}} \alpha_{h^\dagger, i}^{(\ell^*)} \geq \bar{\alpha} \cdot (1 - J_{\text{late}}) \cdot \max_h |\mathcal{R}_h^{(\ell^*)}|, \quad (12)$$

using Eq. (11). Under eviction with noise-retention ratio γ and a top- k scorer that does not a priori distinguish signal-side residuals from noise positions (an additional modelling assumption beyond (A1)–(A4) which we make explicit here), positions in $\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}$ inherit the same effective per-position retention rate γ as the noise set. Those retained for only a subset of heads appear as distractors to the remaining heads after the cache is merged. Conditional on this retention,

$$\mathbb{E}[\delta_t] \geq \bar{\alpha} \cdot (1 - J_{\text{late}}) \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \cdot (1 - \gamma), \quad (13)$$

because the $(1 - \gamma)$ fraction of un-retained positions in the residual private sets no longer balance the heads that did keep them, leaving a residual distractor mass proportional to $(1 - \gamma)$. This is a new bound derived in our surrogate; it is structurally analogous to Bui’s Corollary 3.2 (which relates pre- and post-eviction dilution via $\delta' = (\rho_D/\rho_U)\delta/[(1 - \delta) + (\rho_D/\rho_U)\delta]$), but the input here is the head-disagreement-induced distractor set rather than a pre-existing dilution.

Step 5 (closing the bridge: solve for D). Eq. (13) uses $(1 - \bar{J}_{\text{late}})$, which we relate to the observable D via Eq. (10): $D = \bar{J}_{\text{early}} - \bar{J}_{\text{late}}$. Since $\bar{J}_{\text{early}} \leq 1$ trivially (Jaccard is bounded by one), $D \leq 1 - \bar{J}_{\text{late}}$, so substituting gives

$$\mathbb{E}[\delta_t] \geq \bar{\alpha} \cdot D \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \cdot (1 - \gamma). \quad (14)$$

The substitution $1 - \bar{J}_{\text{late}} \geq D$ holds always; we do not require the stronger $\bar{J}_{\text{early}} \approx 1$ assumption. The looser the value of $\bar{J}_{\text{early}}$, the larger the gap between $1 - \bar{J}_{\text{late}}$ and D , and the looser this bound is relative to its tightest form.

The $\log(1/\gamma)$ factor on the right-hand side of Eq. (7) is a heuristic amplification term, not a rigorous consequence of the steps above. The intended argument is that after eviction the cache retains γN noise positions, and the maximum of γN sub-Gaussian logits concentrates near $\sigma_z \sqrt{2 \log \gamma N}$; the multiplicative penalty on the head signal-to-noise budget should scale with the post-eviction noise floor relative to the pre-eviction floor. A careful Taylor expansion of $\sqrt{2 \log N} - \sqrt{2 \log \gamma N}$ gives a difference of order $\log(1/\gamma)/\sqrt{2 \log N}$, not $\log(1/\gamma)$ itself; the log factor therefore captures only the worst-case noise-tail shift at small N and not the asymptotic regime. We retain the $\log(1/\gamma)$ factor in Eq. (7) as a convenient packaging of the eviction amplification, but we flag this as gap (G5) in the list below. Readers willing to drop the factor entirely can read Eq. (7) as $D \geq c_1$ with $c_1 = 1/J_{\text{early}}$; this is the conservative no-amplification reading, and it is also empirically consistent with the $\tau = 0.07$ threshold we calibrate (Section 5).

Two heuristic observations close the bridge to $(1 - A_{\text{full}})$; both are leading-order estimates rather than rigorous bounds, and we flag them in the gap list. (i) Under (A1) noise-isotropy and the high-SNR regime, the average per-position mass on a relevant position satisfies $\bar{\alpha} \approx \alpha_{\mathcal{R}}/R$, and under (A3) the head-private sets collectively cover most of $\mathcal{R}$ in the dilution-prone regime so that $\max_h |\mathcal{R}_h^{(\ell^*)}| \approx R$. The product is therefore $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \approx \alpha_{\mathcal{R}}$ as an order-of-magnitude estimate. (The set containment $\mathcal{R}_h^{(\ell^*)} \subseteq \mathcal{R}$ alone gives the reverse inequality $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \leq \alpha_{\mathcal{R}}$, so the leading-order substitution is the honest reading and is flagged in the gap list.) (ii) Combining the headroom factorisation of Appendix E (Eq. (4)) with the strictly increasing accuracy surrogate gives $(1 - A_{\text{full}}) \leq C \cdot \alpha_{\mathcal{R}}$ on dilution-prone tasks for a constant C depending only on the slope of $\hat{A}(\cdot)$ near the operating SNR.

Combining (i) (as a leading-order estimate) and (ii) with Eq. (14) yields the operational bridge

$$\mathbb{E}[\delta_t] \gtrsim \frac{(1 - \gamma)}{C} \cdot D \cdot (1 - A_{\text{full}}), \quad (15)$$

where $\gtrsim$ denotes leading-order in the surrogate; the substitution $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \approx \alpha_{\mathcal{R}}$ is the source of the $\gtrsim$ rather than $\geq$. The conservative rigorous reading replaces $\alpha_{\mathcal{R}}$ on the right-hand side with

$\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \leq \alpha_{\mathcal{R}}$ and drops the bridge to $(1 - A_{\text{full}})$ entirely; we use $\gtrsim$ for the operational reading.

Eq. (15) is the load-bearing bridge between the observable D and Bui’s dilution under the empirical headroom proxy. Eq. (7) in Section 6 packages this relationship as a threshold inequality. Substituting $D \geq T$ into Eq. (15) gives $\mathbb{E}[\delta_t] \gtrsim ((1 - \gamma)T/C) \cdot (1 - A_{\text{full}})$, so any chosen threshold T on D produces a corresponding $c_2 = (1 - \gamma)T/C$. Choosing $T = c_1 \log(1/\gamma)$ with $c_1 = 1/J_{\text{early}}$,

$$c_1 = \frac{1}{J_{\text{early}}}, \quad c_2(\gamma) = \frac{(1 - \gamma) \cdot \log(1/\gamma)}{C \cdot J_{\text{early}}}. \quad (16)$$

J_{early} enters the threshold as a model-specific scale that makes c_1 commensurate with the empirical Jaccard floor across heads. On our test models $J_{\text{early}} \in [0.4, 0.7]$, giving $c_1 \in [1.4, 2.5]$; the empirical $\tau = 0.07$ is well below this, consistent with the constants being loose (per the $O(H)$ slack in Step 3 and the heuristic substitution in observation (i) above). Both constants carry an $O(H)$ multiplicative slack from the pairwise-to- H -wise step of Step 3, and the $\log(1/\gamma)$ factor in $c_1 \log(1/\gamma)$ is heuristic per gap (G5). The conservative no-amplification reading drops the $\log(1/\gamma)$ factor and gives the threshold $D \geq c_1 = 1/J_{\text{early}}$ with rate $c_2(\gamma) = (1 - \gamma)/(C \cdot J_{\text{early}})$ directly.

The converse direction (heuristic, sufficient-side only). The partition predictor’s operational claim is the contrapositive: *small $D \Rightarrow$ low dilution $\Rightarrow$ eviction is risky*. We sketch two arguments and flag that neither is a separate necessity proof.

The first is a comment on the sufficient-side lower bound: under (A3) for the capacity-bound regime $\mathcal{C}(\mathcal{R}_h^{(\ell)} = \mathcal{R}$ for all (ℓ, h)), the union and intersection coincide, so the right-hand side of Eq. (13) is zero. Since Eq. (13) is a *lower* bound on $\mathbb{E}[\delta_t]$, a zero right-hand side does not bound δ_t from above; it only says the sufficient lower bound becomes vacuous.

The second is an upper bound on δ_t that does provide some necessary-side content, but only inside the surrogate. Under (A1) with margin m satisfying the threshold of Lemma 1, and (A3) regime $\mathcal{C}$ so that $U_t = \mathcal{R}$ for every head, the pairwise top- k Jaccard between heads is $\geq J_{\text{early}}$ at every layer (no late-layer divergence), so $|\mathcal{U}|/|\mathcal{I}| \rightarrow 1$ and there is no residual distractor mass. The attention mass on $\mathcal{N}$ for any single head is bounded by the sub-Gaussian tail $O(\exp(-m^2/(4\sigma_z^2)))$ by Lemma 1, and Bui’s dilution is $\delta_t \leq O(\exp(-m^2/(4\sigma_z^2)))$ in the surrogate. This is the surrogate-level converse; lifting it to real attention requires both a lower bound on m for capacity-bound tasks (which we have not verified) and the surrogate identification (A4) of U_t with $\mathcal{I}^{(\ell^*)}$. The boxed Eq. (7)/(16) is the sufficient direction only.

Empirical validation. Table 12 reports the mean head-agreement drop D across the five dilution-prone tasks (qa_1, qa_2, vt, niah_multivalued, fwe) and the capacity-bound task (NIAH-MK3) on Qwen 1.5B, Qwen 3B, and Mistral. In every (model, context) cell, NIAH-MK3 has the smallest drop (including the cell where it is negative, Qwen 3B 16K). The *ordering* of tasks by D is preserved across architectures, and the partition between dilution-prone and capacity-bound is exactly the partition predicted by Eq. (15): cells with $D \approx 0$ have $\mathbb{E}[\delta_t]$ bounded by the noise floor, and the per-input recovery rate ρ in Table 1 is ≤ 0.02 on all such cells. The bridge inequality is therefore empirically consistent with both directions on this task suite. We do not promote this to a formal claim because the sample size per cell is $N = 20$ inputs, the constants in Eq. (16) are not tight, and the necessity direction is not proved.

What is *not* proved. We list the gaps so a reviewer reading this appendix sees the scope clearly.

1. **The constants $c_1, c_2(\gamma)$ in Eq. (16) are loose.** The pairwise-to- H -wise step in Step 3 loses tightness: the union-vs-intersection gap can scale with H in the worst case, so c_1 is honest only up to an $O(H)$ multiplicative slack. A tight version would replace the pairwise inequality with a full higher-order union-intersection identity and track the layerwise covariance of head subsets.
2. **The pairwise Jaccard is not the same object as the H -wise union/intersection ratio.** Eq. (11) uses the weaker inequality “some head has a large private residual set”; this is correct but discards information when many heads share the same residual.

3. **Observation (i) in Step 5 is a leading-order estimate, not a rigorous inequality.** The exact reverse inequality $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \leq \alpha_{\mathcal{R}}$ holds (set containment), so the substitution $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \approx \alpha_{\mathcal{R}}$ used in Eq. (15) relies on (A1)+(A3) jointly to push the high-SNR approximation $\max_h |\mathcal{R}_h^{(\ell^*)}| \approx R$. Eq. (15) therefore uses $\gtrsim$ rather than $\geq$.
4. **The identification of Bui’s U_t with the head-common signal set $\mathcal{I}^{(\ell^*)}$ (A4) is a modelling choice, not a derived equality.** Bui’s U_t is defined at the residual-stream level; our $\mathcal{I}^{(\ell^*)}$ is an intersection over per-head attended sets at a single layer. The two coincide in the surrogate by fiat.
5. **The $\log(1/\gamma)$ factor in Step 5 is heuristic.** The sub-Gaussian maximum over γN retained-noise positions shifts noise-tail scale by $O(\log(1/\gamma)/\sqrt{\log N})$, not $\log(1/\gamma)$. We retain the cleaner $\log(1/\gamma)$ packaging in Eq. (7) as a worst-case bound at small N , and we offer the conservative reading $D \geq c_1$ (no log term) as a strict alternative for readers who reject the surrogate-level amplification argument.
6. **The converse direction is conditional on a margin assumption we have not verified.** In Step 5 we used $m \geq \sigma_z \sqrt{4 \log(2H^2 L T^2 / \beta)}$ to get Lemma 1; the surrogate-level converse additionally requires that capacity-bound tasks satisfy this margin condition for the *common* relevant set $\mathcal{R}$. We do not test this on real attention weights; doing so is a clean follow-up.
7. **The converse paragraph above is an upper bound on the sufficient-condition argument, not a separate necessary-condition proof.** Eq. (7) with constants from Eq. (16) is the sufficient direction only; the empirical content of necessity is reported but not formalised.
8. **The assumption that heads share a common $\mathcal{R}$ in capacity-bound tasks (A3, regime C) needs verification.** Our evidence is the small empirical D on NIAH-MK3 across cells; whether the underlying mechanism is actually “all heads agree on the same needle position” or some weaker form of agreement (heads agreeing on a length- $O(1)$ neighborhood of the needle) is open. The difference between these is invisible to top- k Jaccard but matters for the rigorous proof.
9. **Step 4 implicitly assumes the scorer does not distinguish signal-side residual positions from noise positions a priori.** The $(1 - \gamma)$ factor in Eq. (13) applies the noise-retention ratio γ to positions in $\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)} \subseteq \mathcal{R}$, which are signal-side. The retention rate there equals γ only when a top- k scorer is symmetric across signal vs noise on these residuals; under (A1) noise-isotropy this is the leading-order assumption.
10. **The output-projection averaging in Step 4 treats per-head value contributions as additive.** The single-head SNR analysis of Appendix E is lifted to multi-head by summing attention masses across heads. A precise lift would use the value-vector subspace decomposition of the output projection; under non-orthogonal value subspaces this can shift the constants by $O(H)$ in the worst case, not the $O(1)$ we previously claimed.
11. **The assumed accuracy function $\hat{A}(\text{SNR})$ is treated as task-agnostic.** The constant C in Eq. (15) depends on the slope of $\hat{A}$ near the operating SNR; on tasks where $\hat{A}$ is steep (precise retrieval) versus shallow (aggregation), C differs by an order of magnitude.

The sketch is therefore a *structural* proof of the sufficient direction: the form of the bound and the dependence on (H, γ) are derived from the surrogate; the numerical values of the constants, the H -tightness, the $\log(1/\gamma)$ factor, and the necessity direction are flagged as future work.

G CONFIDENCE INTERVALS FOR THE HEADLINE NUMBERS

Tables 19 and 20 report 95% intervals for every headline delta, computed as described in the uncertainty paragraph of Section 7. Full per-cell derivations ship with the released logs.

Table 19: 95% confidence intervals for the 16 gating deltas of Table 2. Cluster bootstrap over inputs: we resample the (task, id) pairs with replacement (10,000 replicates); each input contributes its per-budget paired gated–plain differences, preserving the pairing and the within-input correlation across budgets; intervals are 2.5–97.5 percentiles. All gated outcomes are the exact $\tau = 0.07$ post-hoc reconstruction of Appendix H. $N = 400$ inputs per cell (4 tasks $\times$ 100), $N = 200$ for Qwen2.5-14B. *Every one of the 16 intervals excludes zero*; the smallest lower bound is +0.093 (Qwen2.5-1.5B, SnapKV).

Model	Base evictor	Δ	95% CI	N
Qwen2.5-1.5B	SnapKV	+0.121	[+0.093, +0.151]	400
Qwen2.5-1.5B	H2O	+0.162	[+0.126, +0.199]	400
Qwen2.5-1.5B	StreamingLLM	+0.163	[+0.128, +0.200]	400
Qwen2.5-1.5B	PyramidKV	+0.152	[+0.118, +0.188]	400
Qwen2.5-3B	SnapKV	+0.259	[+0.225, +0.295]	400
Qwen2.5-3B	H2O	+0.303	[+0.261, +0.347]	400
Qwen2.5-3B	StreamingLLM	+0.430	[+0.383, +0.477]	400
Qwen2.5-3B	PyramidKV	+0.329	[+0.286, +0.372]	400
Qwen2.5-14B	SnapKV	+0.143	[+0.108, +0.181]	200
Qwen2.5-14B	H2O	+0.200	[+0.152, +0.250]	200
Qwen2.5-14B	StreamingLLM	+0.250	[+0.190, +0.310]	200
Qwen2.5-14B	PyramidKV	+0.228	[+0.175, +0.285]	200
Mistral-7B	SnapKV	+0.187	[+0.154, +0.220]	400
Mistral-7B	H2O	+0.236	[+0.196, +0.278]	400
Mistral-7B	StreamingLLM	+0.263	[+0.220, +0.305]	400
Mistral-7B	PyramidKV	+0.231	[+0.191, +0.272]	400

Table 20: 95% CIs for the remaining headline deltas. Cross-architecture rows (Table 5) use the four-budget grid and the same cluster bootstrap; 16K rows are the second half of Table 9; the DBTrimKV row is Table 16 ($N = 120$ inputs, budgets {128, 256, 512}). The Mistral NIAH-MK3 showcase entries are Wilson score intervals on $N = 100$. The only headline delta whose interval includes zero is Qwen2.5-3B at 16K, the gate-misfire cell that Section 4.2 already reports as a shrunken effect.

Quantity	Point	95% CI	N
Yi-1.5-9B Δ (SnapKV, 4-budget)	+0.390	[+0.338, +0.444]	200
Llama-3.1-8B Δ (SnapKV, 4-budget)	+0.109	[+0.069, +0.154]	200
Qwen2.5-1.5B 16K Δ	+0.032	[+0.013, +0.053]	200
Qwen2.5-3B 16K Δ	+0.034	[−0.002, +0.071]	200
Qwen2.5-14B 16K Δ	+0.071	[+0.034, +0.107]	120
Mistral-7B 16K Δ	+0.116	[+0.079, +0.156]	200
DBTrimKV wrapper Δ (RULER 4K)	+0.233	[+0.172, +0.297]	120
Mistral 4K NIAH-MK3, gated acc. @ $b=0.0625$	0.89	[0.814, 0.937]	100
Mistral 4K NIAH-MK3, plain acc. @ $b=0.0625$	0.00	[0.000, 0.037]	100

H REPRODUCTION NOTES

Implementation details that matter for exact reproduction. The cache slicing uses explicit `position_ids` to preserve RoPE alignment; the naive attention-mask trick produces a silent position shift on evicted positions. Long-context (16K) runs use SDPA for prefill and switch to eager for the $w \times T$ observation-window scoring pass to avoid the $O(T^2)$ attention-matrix allocation on 80GB. Head agreement uses Jaccard similarity of top-32 key sets per head pair, averaged over pairs, with early/late layer bins $[0, \lfloor L/3 \rfloor)$ and $[\lfloor 2L/3 \rfloor, L)$. Decoding is greedy throughout, so the gated outcome of any input is exactly reproducible from its recorded head-agreement drop plus the plain-eviction and full-KV outcomes; we use this identity to evaluate gate thresholds post-hoc, and one

matrix cell (Qwen2.5-3B 4K SnapKV, executed at $\tau = 0.04$) is reported at $\tau = 0.07$ via this exact re-evaluation. A pipeline audit verifies that eviction at budget 1.0 matches clean prefill-decode token-for-token up to a trailing-period mismatch on one of six audited examples; the released logs include the audit script.

I SHARPENED DEFERRED PROOFS

This appendix upgrades three results that Appendices E and F left as sketches: (i) the leading constant and a two-sided bound for the scaling formula (4); (ii) the necessary (converse) direction of the D -to-dilution bridge; and (iii) the status of the useful-set identification (A4). Each subsection ends with an explicit “What this does and does not establish” paragraph so the paper’s claims can be scoped exactly. Throughout we keep the notation of Appendices E–F: $\alpha_{\mathcal{R}}$ is the pre-eviction signal mass, γ the noise-retention ratio, $\rho_{\mathcal{R}}, \rho_{\mathcal{N}}$ the retained mass fractions, δ_t Bui’s dilution (6), D the head-agreement drop (3), σ_z the pre-softmax logit-noise scale, m the logit margin, H, L, T the head, layer and length counts, and $\hat{A}(\cdot)$ the strictly increasing, $[0, 1]$ -valued, unit-saturating accuracy surrogate.

I.1 ITEM 1: THE EXACT LEADING CONSTANT AND A TWO-SIDED SCALING BOUND

The sketch in Appendix E obtains $\rho_{KV} \leq K(1 - A_{\text{full}})p_{\text{recov}} + K/4$ by a union bound over the K -point budget grid $\mathcal{B} = \{b_1, \dots, b_K\}$ followed by a covariance residual. We show that the grid factor K is an artifact of an unnecessary union step, that the correct leading constant is exactly 1, and that a distribution-free *two-sided* bound holds with explicit endpoints; the only genuinely unresolved quantity is a covariance, for which we give matching, attainable bounds.

Events. We work inside the surrogate of Appendix E, where the per-input improvement event at a single budget b factorises (their displayed identity) into a headroom indicator and a recovery indicator. Fix the prompt distribution and, for a random prompt x , define the two Bernoulli random variables

$$A(x) := \mathbf{1}[\hat{A}(\text{SNR}(b_{\text{max}}, x)) < 1] \quad (\text{headroom}), \quad B(x) := \mathbf{1}[\exists b \in \mathcal{B} : \text{SNR}_{\text{post}}(b, x) > \text{SNR}_{\text{pre}}(x)] \quad (\text{recovery}). \quad (17)$$

The operational recovery rate is $\rho_{KV} = \Pr_x[A(b^*(x), x) > A(b_{\text{max}}, x)]$ with $b^*(x) = \arg \max_b A(b, x)$ (Eq. (1)). Because b^* is the argmax over the grid, “improvement at b^* ” is exactly “improvement at some $b \in \mathcal{B}$,” and the single-budget surrogate factorisation gives

$$\{A(b^*(x), x) > A(b_{\text{max}}, x)\} = \bigcup_{b \in \mathcal{B}} (A(x) \cap B_b(x)) = A(x) \cap \left(\bigcup_{b \in \mathcal{B}} B_b(x) \right) = A(x) \cap B(x), \quad (18)$$

where $B_b(x) = \{\text{SNR}_{\text{post}}(b, x) > \text{SNR}_{\text{pre}}(x)\}$ and $B = \bigcup_b B_b$. The headroom event does not depend on b , so it pulls out of the union: the union over budgets lives *inside* B . We set $p_{\text{recov}} := \Pr_x[B]$ (recovery at the best budget) and note $\Pr_x[A] = 1 - A_{\text{full}}$ by definition of the surrogate headroom.

Theorem 1 (Exact leading constant; two-sided scaling bound). *With the definitions in (17)–(18) and $p_{\text{recov}} := \Pr_x[B]$, the surrogate per-input recovery rate satisfies the exact identity*

$$\rho_{KV} = (1 - A_{\text{full}})p_{\text{recov}} + \text{Cov}_x(A, B). \quad (19)$$

The content here is not the identity (19), which is merely the definition of covariance for two Bernoullis, but the observation that once p_{recov} is defined as the union probability $\Pr_x[\bigcup_b B_b]$ (the quantity b^ actually selects) rather than the single-budget $\max_b \Pr_x[B_b]$, the budget-grid factor K of Appendix E is absorbed into p_{recov} rather than multiplying it; K reappears inside p_{recov} whenever the recovery budgets are near-disjoint (Corollary 1(ii)). The covariance is bounded, by*

$$0 \leq \text{Cov}_x(A, B) \leq \min\{1 - A_{\text{full}}, p_{\text{recov}}\} - (1 - A_{\text{full}})p_{\text{recov}}, \quad (20)$$

where the lower bound requires the association hypothesis of Lemma 2 below, and the upper bound (Fréchet) is unconditional. Equivalently,

$$(1 - A_{\text{full}})p_{\text{recov}} \leq \rho_{KV} \leq \min\{1 - A_{\text{full}}, p_{\text{recov}}\}. \quad (21)$$

The lower bound is attained iff $A \perp B$; the upper bound is attained iff A and B are Fréchet–Hoeffding comonotone (one event a.s. contains the other).

Proof. Identity (19) is the definition of covariance applied to the two Bernoulli variables in (17), using $\rho_{KV} = \Pr_x[A \cap B] = \mathbb{E}_x[AB]$ from (18) and $\mathbb{E}_x[A] = 1 - A_{\text{full}}$, $\mathbb{E}_x[B] = p_{\text{recov}}$. For (20), the upper endpoint is the Fréchet–Hoeffding inequality $\Pr[A \cap B] \leq \min\{\Pr[A], \Pr[B]\}$, valid for any two events; subtracting the product gives the stated covariance bound and the right side of (21). The lower endpoint $\text{Cov}_x(A, B) \geq 0$ is Lemma 2. Attainment: independence gives equality in the lower bound by construction; comonotonicity ($A \subseteq B$ or $B \subseteq A$ a.s.) gives $\Pr[A \cap B] = \min\{\Pr[A], \Pr[B]\}$, the upper endpoint. $\square$

Lemma 2 (Association of headroom and recovery). *Suppose the noise-isotropy assumption (A1) of Appendix E holds, so that both A and B are non-increasing functions of the single scalar driver $\alpha_{\mathcal{R}}(x)$ (equivalently, $\mathbb{E}[A \mid \alpha_{\mathcal{R}}]$ and $\mathbb{E}[B \mid \alpha_{\mathcal{R}}]$ are non-increasing in $\alpha_{\mathcal{R}}$ and monotone in the same direction). Then $\text{Cov}_x(A, B) \geq 0$.*

Proof. If $A = f(\alpha_{\mathcal{R}})$ and $B = g(\alpha_{\mathcal{R}})$ with f, g both non-increasing, Chebyshev’s association (sum) inequality gives $\mathbb{E}[fg] \geq \mathbb{E}[f]\mathbb{E}[g]$, i.e. $\text{Cov} \geq 0$. If A depends on an auxiliary driver ζ independent of $\alpha_{\mathcal{R}}$ (e.g. a parametric-knowledge covariate), write $\text{Cov}(A, B) = \text{Cov}(\mathbb{E}[A \mid \alpha_{\mathcal{R}}], B)$ by the tower rule and $B = g(\alpha_{\mathcal{R}})$; the same monotone-in-a-common-scalar argument applies to the conditional mean, so the sign is preserved. $\square$

The monotonicity of A in $\alpha_{\mathcal{R}}$ is immediate (larger signal mass $\Rightarrow$ larger $\text{SNR}_{\text{pre}} \Rightarrow$ smaller headroom). The monotonicity of B follows from Eq. (8): the attainable SNR gain from removing noise mass shrinks as $\alpha_{\mathcal{R}} \rightarrow 1$ (less noise to remove), so recovery is less likely at larger $\alpha_{\mathcal{R}}$.

Why the grid factor K was spurious (tight vs. loose regime). The apparent K in Appendix E comes from the step $\Pr_x[\bigcup_b B_b] \leq \sum_b \Pr_x[B_b] \leq K \max_b \Pr_x[B_b]$ followed by identifying $\max_b \Pr_x[B_b]$ with p_{recov} . Defining p_{recov} as the union probability $\Pr_x[\bigcup_b B_b]$ instead, which is the quantity the operational $b^*(x)$ actually selects, removes the union step entirely. The overcount incurred by the discarded step is, exactly,

$$\sum_{b \in \mathcal{B}} \Pr_x[B_b] - \Pr_x\left[\bigcup_b B_b\right] = \sum_{j \geq 2} (-1)^{j+1} \sum_{|S|=j, S \subseteq \mathcal{B}} \Pr_x\left[\bigcap_{b \in S} B_b\right] \geq 0, \quad (22)$$

the inclusion–exclusion tail. Its size is what makes the union bound tight or loose:

Corollary 1 (Tightness characterisation of the budget-grid bound). *Let $p^\# := \max_b \Pr_x[B_b]$ be the single-budget recovery probability.*

- (i) (Monotone gain profile $\Rightarrow$ no K factor.) *If the SNR gain $g(b, x) := \rho_{\mathcal{R}}(b, x)^2 / \gamma(b, x)$ is monotone in b for a.e. x (as it is for a calibrated scorer, since keeping more tokens raises both $\rho_{\mathcal{R}} \rightarrow 1$ and $\gamma \rightarrow 1$ so $g \downarrow 1$), then the events $\{B_b\}$ are nested and $p_{\text{recov}} = \Pr_x[\bigcup_b B_b] = p^\#$. The genuine leading factor is 1; the union bound $K p^\#$ overcounts by $(K - 1)p^\#$.*
- (ii) (Disjoint gain profile $\Rightarrow K$ is real.) *If instead the $\{B_b\}$ are pairwise disjoint (each input recovers at exactly one budget) and rare ($K p^\# \ll 1$), then $p_{\text{recov}} = \sum_b \Pr_x[B_b]$ is genuinely $\Theta(K)$ times larger than $p^\#$; recovery at the best budget really is K -fold more frequent than at a fixed budget, and the factor K is not slack but signal.*

In both cases Theorem 1 holds with constant 1 once p_{recov} is the union probability; the two regimes differ only in how much larger p_{recov} is than the single-budget $p^\#$.

Proof. (i) If $g(\cdot, x)$ is monotone then $\{b : g(b, x) > 1\}$ is an interval anchored at the aggressive end of the grid, so $B_b(x) \subseteq B_{b'}(x)$ whenever b' is more aggressive than b ; the B_b are nested and their union equals the loosest single event, of probability $p^\#$. (ii) For disjoint events $\Pr[\bigcup_b B_b] = \sum_b \Pr[B_b]$ exactly; rareness makes this $\approx K p^\#$. $\square$

Because our scorers are calibrated (the top- k retention curve is monotone in b), the operative regime is Corollary 1(i): the honest constant is 1, and the K and $K/4$ terms of Appendix E should be read as a loose fallback that is never binding on our grid.

What this does and does not establish. *Establishes:* inside the deterministic-SNR surrogate, the scaling law holds as the *exact* identity (19) with leading constant precisely 1 (no budget-grid inflation), and the two-sided bound (21) with fully explicit, distribution-free endpoints; the only free quantity is the covariance $\text{Cov}_x(A, B) \in [0, \min\{\Pr A, \Pr B\} - \Pr A \Pr B]$, and both endpoints are attainable, so no tighter distribution-free constant exists. *Does not establish:* (a) the surrogate itself, that accuracy is a deterministic (or fixed-conditional) function of head SNR, is assumed, not proven, so the constant “1” is exact only relative to that surrogate; (b) the covariance is pinned only to an interval, not a value, because it depends on the joint law of headroom and recovery, which is task- and model-specific; Lemma 2 fixes its *sign* (non-negative) under single-driver monotonicity but not its magnitude; (c) when headroom is driven partly by an $\alpha_{\mathcal{R}}$ -independent factor (parametric-knowledge failures, e.g. the QA cells in Table 1), recovery collapses ($p_{\text{recov}} \rightarrow 0$) so both bounds pinch $\rho_{\text{KV}} \rightarrow 0$ despite headroom, consistent with the observed QA exceptions, but a prediction of the sign structure, not a proof about those tasks. The empirically observed ratio band $[0.25, 0.55]$ is thus an estimate of $p_{\text{recov}} + \text{Cov}/(1 - A_{\text{full}})$, not of a universal constant.

I.2 ITEM 2: THE NECESSARY DIRECTION (CAPACITY-BOUND $\Rightarrow$ SMALL D)

Appendix F proves the sufficient direction (large D forces large δ_t) and states that the converse “capacity-bound $\Rightarrow$ small D ” needs “a margin condition we have not verified.” We state that condition precisely and prove the converse under it, in both a deterministic and a probabilistic (contrapositive) form, with explicit margin dependence.

The margin condition, made precise. Recall $z_{h,i}^{(\ell)} = (q_h^{(\ell)})^\top k_i + \eta_{h,i}^{(\ell)}$ with σ_z -sub-Gaussian noise (A2), and the logit margin $m = m_+ - m_-$ (A1). Define the *margin-to-noise ratio* $\kappa := m/\sigma_z$.

Assumption 1 (κ -separation of the capacity-bound relevant set). For an input in the capacity-bound regime $\mathcal{C}$, the deterministic logit gap between every common relevant position and every noise position at the late layer bin exceeds

$$m \geq 2\sigma_z \sqrt{\log(2H^2LT^2/\beta)}, \quad \text{i.e.} \quad \kappa \geq 2\sqrt{\log(2H^2LT^2/\beta)}. \quad (23)$$

This is exactly the threshold of Lemma 1; naming it as an assumption isolates the one quantity the converse needs. It is the precise form of the informal “separation between the relevant-token logit margin and the noise scale.”

Theorem 2 (Capacity-bound $\Rightarrow$ small head-agreement drop). *Adopt the surrogate (A1)–(A3) of Appendix F and Assumption 1. Suppose the input is capacity-bound in the sense that the late-layer heads share a common relevant set: there is a set $\mathcal{R}_*$ with $\mathcal{R}_h^{(\ell)} = \mathcal{R}_*$ for every head h and every layer ℓ in the late bin, and top- k is taken at $k = |\mathcal{R}_*|$. Then, over the pre-softmax noise,*

$$\Pr[a_{\text{late}} = 1] \geq 1 - \beta, \quad \text{hence} \quad \Pr[D \leq a_{\text{early}} - 1 \leq 0] \geq 1 - \beta, \quad (24)$$

and therefore, for every threshold $\tau > 0$,

$$\Pr[D \geq \tau] \leq \beta = 2H^2LT^2 \exp\left(-\frac{m^2}{4\sigma_z^2}\right) = 2H^2LT^2 e^{-\kappa^2/4}. \quad (25)$$

Proof. By Lemma 1, under (23) the top- k set of every late-bin head (ℓ, h) equals its private set $\mathcal{R}_h^{(\ell)}$ jointly with probability $\geq 1 - \beta$; call this event E . On E , since all late-bin heads share $\mathcal{R}_*$, every head’s top- k set is exactly $\mathcal{R}_*$, so every pairwise top- k Jaccard equals 1 at every late-bin layer, whence $a_\ell = 1$ for each late-bin layer and $a_{\text{late}} = 1$. This proves the first claim of (24). Consequently $D = a_{\text{early}} - a_{\text{late}} = a_{\text{early}} - 1 \leq 0$ on E because $a_{\text{early}} \leq 1$ (Jaccard is bounded by 1), giving the second claim. Finally, $\{D \geq \tau\}$ for $\tau > 0$ is disjoint from E (on E , $D \leq 0$), so $\Pr[D \geq \tau] \leq \Pr[E^c] \leq \beta$; substituting the value of β from Lemma 1 yields (25). $\square$

Equation (25) is the explicit margin dependence: the probability that a capacity-bound input produces a drop as large as any fixed τ decays as $e^{-\kappa^2/4}$ in the margin-to-noise ratio. The bound is only informative when $\beta = 2H^2LT^2e^{-\kappa^2/4} < 1$, which for realistic (H, L, T) requires a large margin-to-noise ratio ($\kappa \gtrsim 10$ at $H=16, L=28, T=4096$); this is a genuine knife-edge (Assumption 1 is close to the informal margin cap $m \leq \log T$ of Appendix E), and outside it the guarantee is vacuous. Consistent with this, the measured D on NIAH-MK3 is *small* rather than reliably non-positive: it is mildly positive in most cells of Table 12 (+0.04 to +0.06) and negative only on Qwen2.5-3B. The theorem therefore predicts $\mathbb{E}[D] \approx 0$ at an $O(\beta)$ floor, with sign indeterminate; the small positive values observed mean real margins sit below the high- κ regime, so the theorem constrains the ordering (NIAH-MK3 smallest) more than the sign.

Corollary 2 (Probabilistic converse / contrapositive). *Let $\mathcal{C}_\star$ denote the capacity-bound event of Theorem 2 (late-layer heads share a common relevant set). Under Assumption 1, for any $\tau > 0$*

$$\Pr[\mathcal{C}_\star \wedge D \geq \tau] \leq \beta = 2H^2LT^2e^{-\kappa^2/4}. \quad (26)$$

Equivalently, if $\Pr[D \geq \tau] > 0$ then $\Pr[\mathcal{C}_\star \mid D \geq \tau] \leq \beta / \Pr[D \geq \tau]$: observing a drop $D \geq \tau$ implies the input is not capacity-bound, except on a noise-tail event of probability $\leq \beta$. This is the a-priori guarantee the gate needs: firing on $D \geq \tau$ mis-fires on a genuinely capacity-bound input only with probability $\leq \beta$.

Proof. On $\mathcal{C}_\star$, Theorem 2 gives $D \leq 0 < \tau$ except on E^c ; hence $\mathcal{C}_\star \cap \{D \geq \tau\} \subseteq E^c$ and $\Pr[\mathcal{C}_\star \cap \{D \geq \tau\}] \leq \beta$. The conditional statement is Bayes' rule. $\square$

What this does and does not establish. *Establishes:* under the explicit κ -separation Assumption 1, the converse holds cleanly in the surrogate, capacity-bound inputs have $D \leq 0$ with probability $\geq 1 - \beta$ (Theorem 2), with the failure probability decaying as $e^{-\kappa^2/4}$; and the operational contrapositive (Corollary 2) bounds the gate's capacity-bound mis-fire probability by the same β . This upgrades the informal "margin condition we have not verified" to a named assumption with a proof and a rate. *Does not establish:* (a) that *real* capacity-bound tasks satisfy $\mathcal{C}_\star$, i.e. that late-layer heads literally share one common relevant set rather than merely agreeing on an $O(1)$ neighbourhood of the needle, top- k Jaccard cannot distinguish these, and the difference is invisible to D but would weaken the deterministic $a_{\text{late}} = 1$ step to $a_{\text{late}} = 1 - o(1)$ (the conclusion $\Pr[D \geq \tau] \leq \beta + o(1)$ survives, but $\mathcal{C}_\star$ is then an idealisation); (b) the numerical value of κ on real attention weights, Assumption 1 is a hypothesis about the network, tested here only through its consequence (small D on NIAH-MK3 in every cell of Table 12), not measured directly; (c) a bound on a_{early} from below, so the theorem controls the sign and the upper tail of D but not how negative D becomes. The converse is thus proved *conditionally on the margin assumption and the common-set model of capacity-bound*; both are now explicit rather than hidden.

I.3 ITEM 3: THE A4 USEFUL-SET IDENTIFICATION

Assumption (A4) of Appendix F identifies Bui's useful set U_t (a residual-stream, decoding-step object) with the head-common signal set $\mathcal{I}^{(\ell^*)} = \bigcap_h \mathcal{R}_h^{(\ell^*)}$ (a single-layer attention object), and the text flags it as "a modelling choice, not a derived equality." We show that a clean exact equality cannot be proven, but a *quantitative approximate* identification can: under a named consensus-usefulness assumption plus ε -closeness of heads, the two sets induce dilution values that agree up to $O(\varepsilon)$, with an explicit constant. We therefore replace the "derived equality" language with a proved approximation plus a named residual assumption.

Set-up. At the divergence layer ℓ^* write the head-private top- k sets $\mathcal{R}_h := \mathcal{R}_h^{(\ell^*)}$, their intersection $\mathcal{I} := \bigcap_h \mathcal{R}_h$ and union $\mathcal{U} := \bigcup_h \mathcal{R}_h$. For a set $\mathcal{S}$ define its induced dilution $\delta(\mathcal{S}) := 1 - \sum_{i \in \mathcal{S}} \alpha_{t,i}$, so Bui's is $\delta^{(U)} := \delta(U_t)$ and ours is $\delta^{(I)} := \delta(\mathcal{I})$. Let $\bar{\alpha}_{\max} := \max_i \alpha_{t,i}$.

Assumption 2 (ε -closeness of heads). The late-layer heads are ε -close in their top- k sets: $\max_{h,h'} |\mathcal{R}_h \triangle \mathcal{R}_{h'}| \leq \varepsilon k$, equivalently pairwise Jaccard $\geq (1 - \varepsilon/2)/(1 + \varepsilon/2) = 1 - O(\varepsilon)$.

Assumption 3 (Consensus-usefulness, CUA). The useful set is sandwiched between the head-consensus and head-union sets: $\mathcal{I} \subseteq U_t \subseteq \mathcal{U}$. The right inclusion ($U_t \subseteq \mathcal{U}$) holds whenever routing to the residual stream is through attention only, so a position no head attends to cannot affect

the next-token margin. The left inclusion ($\mathcal{I} \subseteq U_t$) is the substantive content: a position every head attends to is margin-preserving.

Proposition 1 (Quantitative approximate identification). *Under Assumptions 2–3,*

$$|\delta_t^{(U)} - \delta_t^{(I)}| \leq \sum_{i \in \mathcal{U} \setminus \mathcal{I}} \alpha_{t,i} \leq H(H-1)\varepsilon k \bar{\alpha}_{\max} = O(H^2\varepsilon), \quad (27)$$

where the last step uses $\bar{\alpha}_{\max} = \Theta(1/k)$ in the high-SNR softmax. In particular, as $\varepsilon \rightarrow 0$ (heads perfectly agree) the two dilution values coincide: $\delta_t^{(U)} \rightarrow \delta_t^{(I)}$.

Proof. By Assumption 3, $U_t \Delta \mathcal{I} \subseteq \mathcal{U} \setminus \mathcal{I}$, because $\mathcal{I} \subseteq U_t \subseteq \mathcal{U}$ forces every element of $U_t \Delta \mathcal{I} = U_t \setminus \mathcal{I}$ to lie in $\mathcal{U} \setminus \mathcal{I}$. Since $\delta(\mathcal{S})$ is a signed sum of attention masses over $\mathcal{S}$, $|\delta_t^{(U)} - \delta_t^{(I)}| = |\sum_{i \in U_t} \alpha_{t,i} - \sum_{i \in \mathcal{I}} \alpha_{t,i}| \leq \sum_{i \in U_t \Delta \mathcal{I}} \alpha_{t,i} \leq \sum_{i \in \mathcal{U} \setminus \mathcal{I}} \alpha_{t,i}$, the first inequality of (27). For the cardinality bound, for each head h , $|\mathcal{R}_h \setminus \mathcal{I}| \leq \sum_{h' \neq h} |\mathcal{R}_h \setminus \mathcal{R}_{h'}| \leq \sum_{h' \neq h} |\mathcal{R}_h \Delta \mathcal{R}_{h'}| \leq (H-1)\varepsilon k$ by Assumption 2. Since $\mathcal{U} \setminus \mathcal{I} = \bigcup_h (\mathcal{R}_h \setminus \mathcal{I})$, $|\mathcal{U} \setminus \mathcal{I}| \leq \sum_h |\mathcal{R}_h \setminus \mathcal{I}| \leq H(H-1)\varepsilon k$. Bounding each mass by $\bar{\alpha}_{\max}$ gives the middle inequality; $\bar{\alpha}_{\max} = \Theta(1/k)$ gives the $O(H^2\varepsilon)$ order. $\square$

Remark 1 (When the identification fails). The left inclusion of Assumption 3 ($\mathcal{I} \subseteq U_t$) can fail, and this is exactly the case the equality cannot cover: if a *single* specialised head carries the decisive value for the next-token margin, then a position in $\mathcal{R}_{h_0} \setminus \mathcal{I}$ (attended by that one head, absent from the consensus) can be margin-critical, so $U_t \not\subseteq \mathcal{U} \setminus (\mathcal{R}_{h_0} \setminus \mathcal{I})$ and $U_t \neq \mathcal{I}$ even at $\varepsilon = 0$ of the *other* heads. This is the single-decisive-head regime, precisely a capacity-bound, single-needle retrieval situation, where one retrieval head dominates. The approximation of Proposition 1 is therefore reliable in the dilution-prone regime (many heads, diffuse consensus, no single decisive head) and degrades exactly where the paper already declines to apply the bridge (capacity-bound).

What this does and does not establish. *Establishes:* the identification of U_t with $\mathcal{I}^{(\ell^*)}$ is not an exact derived equality, but under the named Assumptions 2–3 the two sets induce *dilution values* that agree up to $O(H^2\varepsilon)$ in attention mass (Proposition 1), with an explicit constant $H(H-1)k\bar{\alpha}_{\max}$; so wherever heads are ε -close, using $\mathcal{I}^{(\ell^*)}$ in place of U_t perturbs Bui’s δ_t by a controlled $O(\varepsilon)$ term rather than by an uncontrolled amount. *Does not establish:* (a) the substantive left inclusion $\mathcal{I} \subseteq U_t$ (CUA); it is named as Assumption 3, justified by the additive-routing picture, and shown in Remark 1 to fail precisely in the single-decisive-head (capacity-bound) regime, so it is an assumption about the model’s read-out, not a theorem; (b) any control of the H^2 prefactor, which is a worst-case union count and is loose when residuals overlap across heads (the same $O(H)$ -slack issue as Step 3 of Appendix F); (c) a bound when ε is $\Theta(1)$ (genuinely divergent heads, the dilution-prone late layers themselves), there the approximation is vacuous and $\mathcal{I}$ and U_t are related only through the CUA sandwich, not quantitatively. The honest scoping is thus: **A4 is a named modelling assumption (CUA), under which we prove an $O(H^2\varepsilon)$ approximate identification of the dilution values**, not a derived equality; the word “derived equality” in Appendix F should be read as this approximation.

J USE OF LARGE LANGUAGE MODELS

Large language models were used as general-purpose assistants during this project: for literature search and triage, for drafting and copy-editing prose, and as coding aids when writing and analyzing the experiment scripts. All research ideas, hypotheses, experimental designs, and claims are the authors’ own; every reported number was produced by the released code on the stated models and data, and the authors verified the experiments and their reporting. LLMs were not used as a source of scientific facts or citations without verification against primary sources.